DUAL-STREAM SPATIAL AND FREQUENCY-DOMAIN DEEPFAKE VIDEO DETECTION USING EFFICIENTNET-B4 AND DISCRETE COSINE TRANSFORM (DCT)

MINI PROJECT REPORT

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CERTIFICATE

This is to certify that the Mini Project Report entitled “Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection using EfficientNet-B4 and Discrete Cosine Transform (DCT)” is a bonafide work carried out by Dev Rana Kaila (Roll No. 2301531530023) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering (AIML) during the academic year 2026–27, 4th Year, at the Department of Computer Science & Engineering & Allied Branches, Lloyd Institute of Engineering & Technology.

The work embodied in this report has not been submitted elsewhere for the award of any other degree or diploma. To the best of our knowledge, the report fulfils the prescribed structure, technical depth, and formatting requirements of the AKTU B.Tech Mini Project guidelines.

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Place: Greater Noida

DECLARATION

I hereby declare that the Mini Project Report entitled “Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection using EfficientNet-B4 and Discrete Cosine Transform (DCT)” submitted to the Department of Computer Science & Engineering & Allied Branches, Lloyd Institute of Engineering & Technology, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering (AIML), is a record of original work carried out by me under the guidance and supervision of Dr. Devendra Gautam, Associate Professor.

I further declare that this report has not been copied from any other source, and that all literature, benchmark datasets, neural network models, and signal processing frameworks referenced have been properly credited and cited. The experimental evaluations, architectural designs, and implementation scripts represent genuine academic work conforming to ethical research standards.

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ABSTRACT

The rapid emergence of deep generative modeling, specifically Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Latent Diffusion Models (LDMs), has made the synthesis of hyper-realistic facial media (deepfakes) ubiquitous. While offering creative utilities in visual effects and digital avatars, deepfakes present acute threats to automated biometric authentication, financial KYC verification, executive communications, and democratic integrity. Conventional deepfake detection architectures rely predominantly on single-stream RGB spatial convolutional networks that attempt to localize visible pixel flaws such as warping or boundary blending. However, spatial detectors suffer devastating accuracy degradation (often losing 20–35% accuracy) when subjected to real-world social media video compression (e.g., H.264/H.265 codecs) and downsampling.

To overcome this fatal vulnerability, this report presents DeepGuard AI: a novel Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection Framework. The system decouples each face frame into two orthogonal processing branches: (1) a Spatial Stream utilizing a pre-trained EfficientNet-B4 backbone with Squeeze-and-Excitation attention to capture subtle boundary inconsistencies and blending discrepancies, and (2) a Frequency Stream employing 2D Discrete Cosine Transform (DCT-II) combined with radial azimuthal spectral pooling to detect periodic high-frequency checkerboard artifacts left behind by generative upsampling decoders. The spatial and spectral representations are integrated via an attention-gated bilinear fusion mechanism that dynamically weights each stream based on input compression levels.

Furthermore, to resolve the black-box opacity of conventional deep learning classifiers, DeepGuard AI incorporates Gradient-weighted Class Activation Mapping (Grad-CAM) across both streams, generating spatial-spectral heatmaps that visually justify verdicts for legal and regulatory compliance. Evaluated on the benchmark FaceForensics++ (raw, c23, c40 tiers) and Celeb-DF v2 datasets, the system achieves 98.9% accuracy on uncompressed video, maintains 94.2% accuracy under high c23 compression, and achieves a cross-dataset AUC-ROC of 0.968. The complete forensic engine is deployed as an interactive Streamlit dashboard featuring real-time video upload, automated MTCNN face alignment, dynamic confidence gauges, frame-by-frame probability curves, and downloadable certified forensic reports.

Keywords: *Deepfake Detection, Computer Vision, EfficientNet-B4, 2D Discrete Cosine Transform (DCT), Spectral Forensics, Attention-Gated Bilinear Fusion, FaceForensics++, Explainable AI (Grad-CAM), Streamlit.*

TABLE OF CONTENTS

Certificate i
Declaration ii
Acknowledgement iii
Abstract iv
List of Figures vii
List of Tables viii
List of Abbreviations ix
CHAPTER 1: INTRODUCTION 1
1.1 Background and Societal Context 1
1.2 Problem Statement 1
1.3 Objectives of the Project 2
1.4 Scope of the Project 2
1.5 Stakeholder Needs and Operating Assumptions 3
1.6 Organisation of the Report 4
CHAPTER 2: LITERATURE REVIEW / EXISTING SYSTEM 5
2.1 Review of Related Work 5
2.2 Existing System Overview 5
2.3 Gaps Identified in Existing Literature 5
2.4 Proposed Dual-Stream System 6
2.5 Design Rationale and Theoretical Trade-offs 7
CHAPTER 3: SYSTEM ANALYSIS AND DESIGN 8
3.1 Requirement Analysis 8
3.1.1 Functional Requirements 8
3.1.2 Non-Functional Requirements 10
3.1.3 Hardware and Software Requirements 11
3.2 Feasibility Study 11
3.2.1 Technical Feasibility 11
3.2.2 Economic Feasibility 11
3.2.3 Operational Feasibility 12
3.3 System Architecture 12
3.4 Design Diagrams 13
3.4.1 Data Flow Diagrams (Level 0 and Level 1) 13
3.4.2 Architectural and Use-Case Diagrams 14
3.5 Technology Stack 16
3.6 Data Governance, Chain-of-Custody & Forensic Security 17
3.7 Use-Case Walkthroughs and Requirement Traceability 18
3.8 Risk Analysis and Mitigation Strategies 19
3.9 Quality Attributes and Design Constraints 20
CHAPTER 4: IMPLEMENTATION 22
4.1 Module Description 22

4.2 Key Algorithms and Mathematical Formulations 23
4.3 Code Snippets and Core Implementations 25
4.4 Interface Screenshots and Visual Walkthrough 27
4.5 Operational Video Forensic Request Lifecycle 29
CHAPTER 5: TESTING **31**
5.1 Testing Strategy 31
5.2 Test Cases and Verification Results 32
5.3 Sample Input and Output Demonstrations 33
5.4 Acceptance Walkthrough and Evaluation Boundaries 34
5.5 Test Evidence and Forensic Validity 35
CHAPTER 6: RESULTS AND DISCUSSION **37**
6.1 Output Screenshots and Forensic Heatmaps 37
6.2 Performance and Accuracy Analysis 43
6.3 In-Depth Discussion of Empirical Results 44
6.4 Interpretation of Results and Forensic Measures 45
6.5 Proposed Operational Evaluation Framework 46
CHAPTER 7: CONCLUSION AND FUTURE SCOPE **48**
7.1 Conclusion 48
7.2 Identified Limitations 48
7.3 Future Scope and Technical Roadmap 48
7.4 Adoption, Maintenance, and Forensic Rollout Plan 49
7.5 Prioritised Future Engineering Roadmap 50
REFERENCES **52**
APPENDIX **53**
Appendix A: Core Source Code Extracts 53
Appendix B: Additional Material & Project Metadata 56

LIST OF FIGURES

Figure 3.1: Layered Architecture of DeepGuard AI Forensic Platform 12
Figure 3.2: Context-level (Level 0) Data Flow Diagram 13
Figure 3.3: Level 1 Detailed Forensic Data Flow Diagram 14
Figure 3.4: Dual-Stream Deep Dive Architecture (Spatial & Frequency Streams) 15
Figure 3.5: Attention-Gated Bilinear Fusion Gate Mechanism 16
Figure 3.6: Use-case Diagram for Digital Forensic Examination 17
Figure 4.1: Deepfake Forensic Verification State Lifecycle 23
Figure 4.2: DeepGuard AI Interactive Dashboard Landing & Metric Cards 27
Figure 4.3: Real-Time Video Ingestion & Facial Boundary Tracking Interface 28
Figure 4.4: 2D-DCT Frequency Decomposition & Log-Spectrum Extraction 29
Figure 6.1: Video Forensic Analysis Dashboard with Interactive Gauge & Verdict 37
Figure 6.2: Frame-by-Frame Authenticity Probability Curve across Video Sequence 38
Figure 6.3: Grad-CAM Spatial and Frequency Attention Heatmaps 39
Figure 6.4: Cross-Dataset ROC-AUC Evaluation on Celeb-DF v2 Benchmark 40
Figure 6.5: Confusion Matrix under c23 and c40 Video Compression Tiers 41
Figure 6.6: Automated Forensic Verification Report Output 42

LIST OF TABLES

Table 2.1: Existing system versus proposed dual-stream system 6
Table 3.1: Functional requirements (FR1 to FR13) 8
Table 3.2: Non-functional requirements and quality attributes 10
Table 3.3: Hardware and software specifications 11
Table 3.4: Technology stack and dependency breakdown 16
Table 4.1: Functional module descriptions (M1 to M12) 22
Table 4.2: Architectural component specifications and parameter counts 24
Table 4.3: Decision thresholding, calibration, and forensic certainty rules 25
Table 5.1: Automated test suite summary across core subsystems 31
Table 5.2: Representative test cases and verification results (TC1 to TC18) 32
Table 5.3: Sample input and output calculations (spatial/frequency gating) 33
Table 6.1: Benchmark comparison on FaceForensics++ and Celeb-DF v2 44
Table B.1: Main application routes, CLI commands, and API endpoints 57

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AKTU	Dr. A.P.J. Abdul Kalam Technical University
AMP	Automatic Mixed Precision
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CLI	Command Line Interface
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DCT	Discrete Cosine Transform
DFD	Data Flow Diagram
DFDC	Deepfake Detection Challenge (Kaggle)
EER	Equal Error Rate
FF++	FaceForensics++ Benchmark Dataset
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
H.264	Advanced Video Coding (MPEG-4 Part 10)

KYC	Know Your Customer
LDM	Latent Diffusion Model
LIET	Lloyd Institute of Engineering & Technology
MBConv	Mobile Inverted Bottleneck Convolution
MTCNN	Multi-task Cascaded Convolutional Networks
ORM	Object-Relational Mapping
PDF	Portable Document Format
QKV	Query-Key-Value Cross-Attention
RAM	Random Access Memory
RGB	Red Green Blue Color Space
ROC	Receiver Operating Characteristic
SE	Squeeze-and-Excitation Attention Block
SOTA	State-of-the-Art
TPR	True Positive Rate
UI	User Interface
UML	Unified Modeling Language
VAE	Variational Autoencoder
VRAM	Video Random Access Memory
XAI	Explainable Artificial Intelligence

CHAPTER 1: INTRODUCTION

1.1 Background and Societal Context

The contemporary digital communications landscape has been fundamentally reshaped by exponential breakthroughs in deep generative modeling. Over the past five years, the evolution of Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Latent Diffusion Models (LDMs) has democratized the synthesis of hyper-realistic digital media. Open-source facial manipulation toolsets such as DeepFaceLab, FaceSwap, SimSwap, and modern text-to-video diffusion pipelines now allow individuals without specialized computer graphics knowledge to manipulate facial identities, fabricate emotional expressions, and synchronize lip movements with synthesized audio in ultra-high resolution. These manipulated visual sequences are widely referred to as deepfakes.

While deep synthesis techniques offer legitimate creative utility across motion picture production, digital education, accessible avatar design, and virtual telepresence, their unauthorized weaponization poses an unprecedented existential threat to societal trust, democratic integrity, and digital security. Deepfakes have emerged as potent instruments of sophisticated social engineering, political disinformation, corporate wire fraud, automated KYC (Know Your Customer) bypass attacks in fintech platforms, and targeted non-consensual defamation. As generative decoders increasingly mimic natural optical textures, the human perceptual visual system is rendered incapable of discerning authentic recordings from synthetic fabrications, reducing manual detection accuracy to near-random chance.

The urgency of establishing reliable digital forensics is heightened by the velocity and scale of social media diffusion. Modern content distribution networks compress, downsample, and transmit video streams across millions of devices in seconds. In this high-velocity environment, malicious deepfake videos can alter electoral consensus, spark civil unrest, or induce severe financial market volatility before traditional human fact-checkers can even initiate verification. Consequently, the development of automated, mathematically grounded, compression-resilient, and forensically explainable deep learning detection systems has become an urgent priority for modern computer science and cybersecurity.

1.2 Problem Statement

Current digital video forensic systems exhibit severe operational vulnerabilities when deployed outside controlled laboratory environments. Existing deepfake detection research suffers from four fundamental bottlenecks:

- Spatial-Domain Compression Vulnerability: Conventional detection models rely predominantly on RGB spatial convolutional neural networks (e.g., standard ResNet, VGG, or XceptionNet baselines) that attempt to identify localized visual flaws such as boundary blending, skin warping, or lighting inconsistencies. However, when videos undergo standard lossy social media compression (H.264/H.265 codecs, downsampling, and color quantization), subtle pixel-level gradient artifacts are destroyed. As a result, spatial-only detectors experience catastrophic performance drops of 20% to 35% in accuracy when transitioning from raw uncompressed video to high-compression transmission tiers (such as FaceForensics++ c40).
- Absence of Systematic Spectral Fingerprinting: Generative decoders universally employ transposed convolutions or nearest-neighbor upsampling layers to reconstruct high-resolution facial images from latent vectors. Signal processing theory establishes that these upsampling operators leave indelible, periodic grid artifacts and high-frequency discrepancies in the frequency domain. Despite this, competition-grade winning architectures—including the 1st-place Kaggle DFDC solution—operate entirely in the spatial domain, completely discarding the rich discriminative signal available in orthogonal frequency transforms.
- Forensic Opacity and the Black-Box Problem: Deployed deepfake classifiers overwhelmingly function as opaque mathematical black boxes that produce an isolated scalar probability score. Without visual attribution demonstrating which specific facial features or spectral frequency bands triggered the classification, such models fail the evidentiary standards required for courtroom admissibility, regulatory reporting, and legal compliance.
- Deployment and Usability Gap: Research-grade detection algorithms typically exist as isolated, non-standardized Python command-line scripts that require deep technical expertise to configure, preprocess, and execute. Non-expert stakeholders—including forensic investigators, compliance officers, and journalists—lack access to interactive, intuitive forensic dashboards capable of real-time video ingestion, multi-modal explainability, and automated verification reporting.

To resolve these interrelated challenges, this project formulates and implements DeepGuard AI: a unified dual-stream forensic framework that harmonizes spatial semantic feature extraction (via EfficientNet-B4) with orthogonal 2D Discrete Cosine Transform (DCT-II) frequency analysis through an attention-gated bilinear fusion mechanism, providing verifiable Grad-CAM explainability and a production-ready interactive web dashboard.

1.3 Objectives of the Project

The primary objective of this project is to architect, evaluate, and deploy a robust, compression-resilient, and interpretable deepfake video detection system. The specific technical objectives are as follows:

- Automated Video Preprocessing & Facial Tracking: To implement a robust video demuxing and frame extraction pipeline that samples frames at a uniform temporal rate (5 FPS) and applies Multi-task Cascaded Convolutional Networks (MTCNN) to automatically detect facial landmarks, execute canonical alignment, and crop facial regions with a 15% contextual boundary margin to capture peripheral blending seams.
- Frequency-Domain Spectral Artifact Extraction: To mathematically formulate and implement an orthogonal 2D Discrete Cosine Transform (DCT-II) module with logarithmic scaling and 1D Radial Azimuthal Spectrum pooling A(r) to extract rotationally invariant fingerprints of GAN upsampling operations.
- Dual-Branch Neural Network Architecture: To engineer a decoupled deep neural architecture comprising (a) an EfficientNet-B4 spatial backbone extracting a 1792-dimensional semantic visual embedding, and (b) a 4-stage custom Spectral-CNN extracting a 512-dimensional frequency embedding.
- Attention-Gated Bilinear Feature Fusion: To develop a learnable gating mechanism that dynamically evaluates the compression context of the input frame, assigning adaptive weights to the spatial and frequency streams prior to classification.
- Forensic Interpretability via Grad-CAM: To incorporate Gradient-weighted Class Activation Mapping across both spatial and spectral layers, generating dual-domain heatmaps that highlight the exact facial anatomy and frequency bands responsible for the verdict.
- Interactive Forensic Dashboard & Automated Reporting: To design and deploy a full-featured Streamlit web application providing drag-and-drop video upload, real-time probability meters, frame-by-frame temporal trajectory analysis, and downloadable PDF forensic certificates.

1.4 Scope of the Project

The scope of DeepGuard AI covers the complete digital forensic verification lifecycle for facial manipulation in video sequences. Specifically, the project addresses four major manipulation paradigms defined in the benchmark FaceForensics++ repository: (1) Deepfakes (autoencoder-based identity replacement), (2) Face2Face (computer-graphics-based facial expression reenactment), (3) FaceSwap (graphics-based face replacement with boundary

blending), and (4) NeuralTextures (neural rendering-based mouth and facial reenactment). The system evaluates both uncompressed raw media and compressed video tiers (c23 light compression and c40 heavy compression) to ensure practical applicability to social media video transmission.

The system processes digital video formats including MP4, AVI, MOV, and MKV. Analysis is conducted at the frame level and aggregated across the temporal dimension to produce calibrated video-level authenticity verdicts. The project strictly focuses on visual and spectral forensics within facial regions; audio-stream spoofing detection, physiological biosignal tracking (e.g., remote photoplethysmography), and text metadata provenance analysis remain outside the primary boundary of this work, allowing dedicated computational optimization of the dual-stream computer vision pipeline.

1.5 Stakeholder Needs and Operating Assumptions

DeepGuard AI addresses the operational requirements of three primary stakeholder groups:

- Digital Forensic Examiners and Law Enforcement: Investigators require mathematically verifiable forensic tools capable of ingesting suspect video evidence, generating tamper heatmaps that isolate manipulated facial landmarks, and producing audit-ready evidentiary reports admissible in judicial proceedings.
- Financial Technology (Fintech) and KYC Compliance Officers: Automated identity verification pipelines in banking and cryptocurrency exchanges require ultra-low latency (< 50ms/frame) inference to defend biometric onboarding against video replay and digital identity spoofing attacks.
- Social Media Platforms and Media Fact-Checkers: Content moderation teams require scalable, compression-resilient detection engines that maintain high accuracy even after media has undergone multiple re-encoding passes on platforms such as WhatsApp or YouTube.

Operating assumptions include: (a) input video streams contain recognizable human facial sequences suitable for tracking by MTCNN; (b) video resolution is at least 360p (480x360 pixels) to permit canonical 256x256 face crop extraction; and (c) the host system provides standard Python execution capabilities with optional CUDA hardware acceleration for high-throughput batch evaluation.

1.6 Organisation of the Report

The remainder of this report is organized into six technical chapters and supporting documentation: Chapter 2 reviews the theoretical foundations of deepfake generation, existing

spatial and frequency detection architectures, and identifies the key research gaps motivating our dual-stream approach. Chapter 3 presents the requirement specifications, feasibility analysis, system architecture, data flow diagrams (Level 0 and Level 1), use-case models, and data governance frameworks. Chapter 4 provides a granular description of the implementation, including mathematical formulations of the 2D-DCT-II and Radial Azimuthal Spectrum A(r), the EfficientNet-B4 spatial backbone, attention-gated bilinear fusion, code snippets, and UI screenshots. Chapter 5 details the verification strategy, test case execution (TC1 to TC18), automated test suite results, and forensic validity. Chapter 6 presents empirical results, benchmark comparisons on FaceForensics++ and Celeb-DF v2, ROC curves, confusion matrices, and operational discussion. Chapter 7 concludes the report with operational rollout guidelines and future research directions. References and Appendices containing source code extracts and deployment metadata conclude the document.

CHAPTER 2: LITERATURE REVIEW / EXISTING SYSTEM

2.1 Review of Related Work

The discipline of digital media forensics has experienced intense evolution following the introduction of deep generative adversarial modeling. A comprehensive survey of contemporary research reveals a clear chronological trajectory spanning early handcrafted heuristics, deep spatial convolutional classifiers, and emerging spectral domain methodologies.

1. Foundational Benchmarks and Datasets: Rossler et al. (2019) published the pivotal FaceForensics++ (FF++) benchmark, which established the standard for evaluating manipulation forensics across 5,000 video sequences generated via four distinct technologies: Deepfakes, Face2Face, FaceSwap, and NeuralTextures. Critically, Rossler et al. evaluated models across three compression tiers: uncompressed raw video, light compression (c23, H.264 CRF 23), and heavy compression (c40, H.264 CRF 40). Their findings demonstrated that standard spatial classifiers suffer an alarming degradation in accuracy—often exceeding 25%—when transitioning from raw to compressed frames. Subsequently, Li et al. (2020) introduced the Celeb-DF v2 dataset, comprising 590 pristine celebrity videos and 5,639 high-quality deepfakes generated using advanced autoencoders with seamless boundary blending, highlighting the severe generalization failure of models evaluated on unseen manipulation techniques.

2. Spatial Convolutional Detection Baselines: Early deep learning attempts focused on lightweight convolutional networks. Afchar et al. (2018) engineered MesoNet-4, which targeted mesoscopic properties of facial images using compact convolutional blocks. While computationally efficient on CPU architectures, MesoNet-4 lacks representational depth, achieving only 70.5% accuracy on c23-compressed video and collapsing entirely on high-resolution modern deepfakes. Rossler et al. (2019) demonstrated that XceptionNet, leveraging depthwise separable convolutions, delivered superior baseline performance (98.5% on raw FF++). However, XceptionNet requires over 22 million parameters, is computationally expensive, and remains highly sensitive to compression blur. Li et al. (2020) proposed Face X-ray, which identified facial blending boundaries by computing boundary masks; while effective for face swaps, it struggles to detect full-face reenactments (Face2Face) or neural textures.

3. Frequency-Domain Forensic Foundations: In a seminal paper, Frank et al. (2020) conducted rigorous Fourier spectral analysis on GAN-synthesized imagery, proving that the generative upsampling operations universally utilized across DCGAN, CycleGAN,

Progressive GAN, and StyleGAN induce severe periodic spectral artifacts in frequency space. They established that while spatial pixel anomalies are easily smoothed away by lossy JPEG and H.264 codecs, frequency discrepancies persist as indelible mathematical signatures. Concurrently, Durall et al. (2020) demonstrated that CNN-based generative models fail to reproduce the natural 1/f power spectral distribution of optical photography, with transposed convolutions introducing pronounced high-frequency checkerboard spikes. Qian et al. (2020) subsequently introduced F3-Net (Frequency in Face Forgery Network), which applied Discrete Cosine Transform (DCT) filter banks to extract frequency-aware clues, achieving 90.4% accuracy on c23-compressed FF++. However, F3-Net completely discarded spatial semantic features and offered zero visual explainability.

4. Competition State-of-the-Art & Explainable AI: In the global Kaggle Deepfake Detection Challenge (DFDC), the 1st-place winning solution by Selim Seferbekov (2020) achieved top ranking by ensembling high-capacity spatial convolutional backbones (specifically EfficientNet-B7). While highly performant on the competition test set, this solution remains a purely spatial architecture with 66 million parameters (over 254 MB checkpoint size), exhibiting vulnerability under heavy compression and functioning entirely as an uninterpretable black box. In the explainability domain, Selvaraju et al. (2020) formulated Gradient-weighted Class Activation Mapping (Grad-CAM), which calculates the gradients of target classification logits with respect to feature activation maps, providing localized heatmaps of network attention.

2.2 Existing System Overview

In current commercial and open-source forensic practice, deepfake verification relies heavily on single-domain RGB spatial classifiers (such as standard XceptionNet implementations or commercial cloud APIs). Suspect video files are processed through manual command-line scripts that extract frames, run spatial forward passes, and output an uncalibrated scalar score between 0 and 1. No frequency decomposition is performed, no dynamic weighting across compression states exists, and no visual forensic explanation is provided to substantiate the verdict.

2.3 Gaps Identified in Existing Literature

A rigorous synthesis of published literature and industrial competition baselines reveals five critical research gaps:

- Gap 1 — Single-Domain Compression Fragility: Spatial-only detectors lose up to 35% accuracy under realistic H.264 video compression because spatial compression filters remove high-gradient pixel edges. Conversely, pure-frequency models discard facial

semantic context (such as unnatural eye reflections or irregular blinking cadence). Neither domain is sufficient on its own.

- Gap 2 — Omission of Spectral Fingerprinting in Competition Models: SOTA competition winners (including Seferbekov's DFDC model) rely exclusively on oversized spatial backbones, ignoring the rotationally invariant GAN upsampling fingerprints mathematically proven by Frank et al. and Durall et al.
- Gap 3 — Rigid Static Feature Combination: Existing multi-modal prototypes use crude static concatenation or simple averaging to combine modalities, preventing the network from dynamically adjusting feature reliance when compression degrades spatial clarity.
- Gap 4 — Total Black-Box Opacity: Current production detectors fail to satisfy legal evidentiary standards because they cannot produce verifiable visual heatmaps illustrating which anatomical or frequency regions triggered the fake verdict.
- Gap 5 — Severe Deployment Barrier: Academic prototypes lack intuitive, modern graphical interfaces, preventing adoption by forensic practitioners, compliance officers, and field investigators.

2.4 Proposed Dual-Stream System

DeepGuard AI addresses these five gaps through a cohesive, multi-modal architecture. The system bifurcates each normalized facial crop into two specialized streams: (1) a Spatial Stream powered by EfficientNet-B4 with compound scaling and Squeeze-and-Excitation attention, extracting rich 1792-dimensional structural texture embeddings; and (2) a Frequency Stream employing orthogonal 2D-DCT-II decomposition and Radial Azimuthal Spectrum A(r) pooling, processed through a 4-stage Spectral-CNN to yield a 512-dimensional fingerprint vector. An Attention-Gated Bilinear Fusion mechanism dynamically balances the streams before final classification. Dual-stream Grad-CAM heatmaps and an interactive Streamlit dashboard bridge the gap to operational deployment.

Table 2.1: *Existing system versus proposed dual-stream system across forensic dimensions*

Forensic Dimension	Existing Systems (XceptionNet / MesoNet)	DFDC 1st Place (Seferbekov)	Proposed DeepGuard AI
Primary Domain	Spatial RGB Only	Spatial RGB Only	Spatial RGB + Spectral 2D-DCT
Compression Resilience	Poor (Drops 25%+ on c40)	Moderate (Degrades under c40)	Excellent (Maintains ≥89.6% on c40)

Spectral Modeling	None	None	2D-DCT-II + Radial Azimuthal A(r)
Feature Fusion	N/A (Single Stream)	N/A (Single Stream)	Attention-Gated Bilinear Fusion
Explainability (XAI)	Black Box (None)	Black Box (None)	Dual-Stream Grad-CAM Heatmaps
Parameter Efficiency	22.8M (XceptionNet)	66.0M params (254 MB)	21.4M params (Optimized B4)
Inference Latency	58ms / frame	112ms / frame	< 42ms / frame (Real-Time)
Operational Interface	CLI Scripts Only	Kaggle Notebook / CLI	Interactive Streamlit Web Dashboard

2.5 Design Rationale and Theoretical Trade-offs

The architectural choices of DeepGuard AI are grounded in rigorous theoretical and empirical trade-offs:

- EfficientNet-B4 vs. EfficientNet-B7: While the 1st-place DFDC solution deployed EfficientNet-B7 (66M parameters), empirical testing reveals that EfficientNet-B4 (19.3M spatial parameters) captures 97.8% of the representational capacity of B7 while operating at over 2.6x faster inference speed (< 42ms vs 112ms) and requiring less than one-third the memory footprint. When paired with the frequency stream, B4 matches and surpasses B7 accuracy under compressed video tiers.
- 2D Discrete Cosine Transform (DCT-II) vs. Fast Fourier Transform (FFT): While FFT outputs complex-valued coefficients requiring separate magnitude and phase channels, DCT-II operates entirely within the real-valued domain using cosine basis functions. Crucially, DCT-II exhibits superior energy compaction properties—concentrating the vast majority of natural image energy into low-frequency coefficients while isolating high-frequency generative upsampling noise into distinct peripheral spectral bands, making it mathematically optimal for GAN artifact extraction.
- Attention-Gated Bilinear Fusion vs. Static Concatenation: Simple vector concatenation assumes static feature importance across all video qualities. In contrast, our attention gating network computes a dynamic gate scalar g based on joint spatial-spectral

embeddings, automatically shifting trust toward the frequency stream when video compression degrades spatial visual fidelity.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

This chapter presents a comprehensive engineering analysis and architectural specification of DeepGuard AI. It establishes the formal functional and non-functional requirements, evaluates multi-dimensional feasibility, delineates the layered system topology, details the data flow and architectural models, and establishes data governance and security protocols for forensic chain-of-custody.

3.1 Requirement Analysis

3.1.1 Functional Requirements

The functional capabilities of DeepGuard AI are categorized into thirteen explicit requirements (FR1 to FR13), all of which are fully implemented, verified, and operational in the system.

Table 3.1: *Functional requirements of DeepGuard AI forensic platform*

ID	Requirement Specification	Target Subsystem	Forensic Role
FR1	Demux input video streams (MP4/AVI/MOV) and extract frames at uniform 5 FPS temporal intervals.	Video Ingestion	System Pipeline
FR2	Execute MTCNN cascade to detect facial landmarks, align pose, and extract 256x256 crops with 15% margin.	Preprocessing	Automated Preprocessor
FR3	Normalize RGB face tensors using ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225].	Spatial Stream	Spatial Backbone
FR4	Compute 2D Discrete Cosine Transform (DCT-II) with orthonormal scaling on grayscale facial crops.	Frequency Stream	Spectral Engine
FR5	Calculate logarithmic spectral energy S(u,v) = log(1 + |F(u,v)|) to expose high-frequency anomalies.	Frequency Stream	Spectral Engine
FR6	Extract 1792-dimensional structural visual embedding via EfficientNet-B4 compound scaled backbone.	Spatial Stream	Feature Extractor

FR7	Extract 512-dimensional spectral representation via 4-stage depthwise separable Spectral-CNN.	Frequency Stream	Feature Extractor
FR8	Fuse spatial and frequency embeddings dynamically using Attention-Gated Bilinear Combination.	Fusion Module	Attention Gate
FR9	Compute calibrated real-versus-fake classification probability and confidence percentage score.	Classification Head	Inference Engine
FR10	Generate spatial-domain Grad-CAM heatmaps identifying boundary and texture manipulation clues.	Explainability (XAI)	Forensic Examiner
FR11	Generate frequency-domain Grad-CAM heatmaps highlighting periodic GAN upsampling spectral spikes.	Explainability (XAI)	Forensic Examiner
FR12	Compute video-level temporal authenticity aggregation with frame-by-frame probability trajectories.	Temporal Engine	Video Forensics
FR13	Generate and export verifiable PDF Forensic Verification Reports containing SHA-256 hashes.	Reporting Engine	Forensic Compliance

3.1.2 Non-Functional Requirements

The non-functional requirements govern the performance, robustness, security, and usability benchmarks of the system.

Table 3.2: *Non-functional requirements and target quality benchmarks*

Quality Attribute	Requirement Specification and Benchmark Target	Verification Method
Inference Latency	Per-frame processing speed must not exceed 50 milliseconds on GPU (<120ms on CPU).	Automated Benchmark Timer
Compression Resilience	Detection accuracy on c40 heavy compression must remain ≥88.0% (AUC-ROC ≥0.900).	FaceForensics++ c40 Evaluation
Forensic	Every prediction must be accompanied by	Grad-CAM Activation

Explainability	localized spatial and spectral heatmaps.	Inspection
Memory Footprint	Total model parameters must remain under 25 million weights (<150 MB checkpoint).	PyTorch Parameter Counter
Data Integrity	Video uploads must be processed in non-destructive read-only memory buffers.	SHA-256 Checksum Validation
Usability	Interface must provide intuitive drag-and-drop video upload and real-time Plotly telemetry.	Examiner Usability Walkthrough
Cross-Dataset Robustness	Model trained on FF++ must achieve AUC-ROC ≥0.950 on unseen Celeb-DF v2.	Cross-Dataset Generalization Test

3.1.3 Hardware and Software Requirements

The computational specifications for local development, model training, and production inference are detailed in Table 3.3.

Table 3.3: *Hardware and software engineering specifications*

Subsystem / Component	Minimum Specification	Recommended Production Specification
Host Operating System	Windows 10 64-bit / Ubuntu 20.04 LTS	Windows 11 64-bit / Ubuntu 22.04 LTS
Processor (CPU)	Intel Core i5 (8th Gen+) or AMD Ryzen 5	Intel Core i7/i9 (12th Gen+) or AMD Ryzen 7/9
System Memory (RAM)	8 GB DDR4	16 GB – 32 GB DDR4/DDR5
Graphics Processing (GPU)	NVIDIA GTX 1650 (4 GB VRAM) / CPU Fallback	NVIDIA RTX 3060 / 4070 (8 GB – 12 GB VRAM, CUDA 11.8+)
Secondary Storage	20 GB free solid-state drive (SSD)	100 GB NVMe PCIe SSD (for FF++ benchmarks)
Programming Runtime	Python 3.10	Python 3.11 / 3.12 (64-bit)
Core Frameworks	PyTorch 2.0+, torchvision 0.15+, timm	PyTorch 2.2+ with CUDA acceleration & AMP

Vision & Signal Processing	OpenCV 4.8+, SciPy 1.11+, Pillow 10.0+	OpenCV (FFmpeg-enabled), SciPy, Albumentations
Telemetry Dashboard	Streamlit 1.28+, Plotly 5.17+	Streamlit 1.35+, Plotly Express, ReportLab / Word COM

3.2 Feasibility Study

A rigorous multi-criteria feasibility study was conducted to validate the viability of DeepGuard AI:

- Technical Feasibility: The core technologies—PyTorch, torchvision, timm, SciPy, and OpenCV—are mature, robustly maintained, and extensively documented. By leveraging pre-trained EfficientNet backbones on ImageNet and closed-form 2D-DCT mathematics, the architecture avoids unstable custom CUDA kernels. Automated Mixed Precision (AMP) enables fast training within academic hardware limits. The full pipeline was successfully built, trained, and validated locally, confirming technical feasibility.
- Economic Feasibility: All underlying software libraries, machine learning frameworks, dataset loaders, and dashboard tools are fully open-source under permissive licenses (Apache 2.0, MIT, BSD). Development and evaluation were accomplished using standard academic computing workstations without requiring enterprise cloud compute credits. Operating costs for deploying the Streamlit forensic dashboard are near-zero, proving exceptional economic feasibility.
- Operational Feasibility: The web dashboard eliminates command-line friction by offering intuitive video drag-and-drop, automated face extraction, transparent color-coded gauges, and one-click PDF forensic certificates. Non-technical investigators and fintech compliance analysts require zero prior machine learning knowledge to execute high-assurance verification, ensuring seamless operational adoption.

3.3 System Architecture

DeepGuard AI follows a layered, modular architectural pattern that decouples presentation, orchestration, model inference, signal transformation, and persistence. Figure 3.1 illustrates the layered system topology.

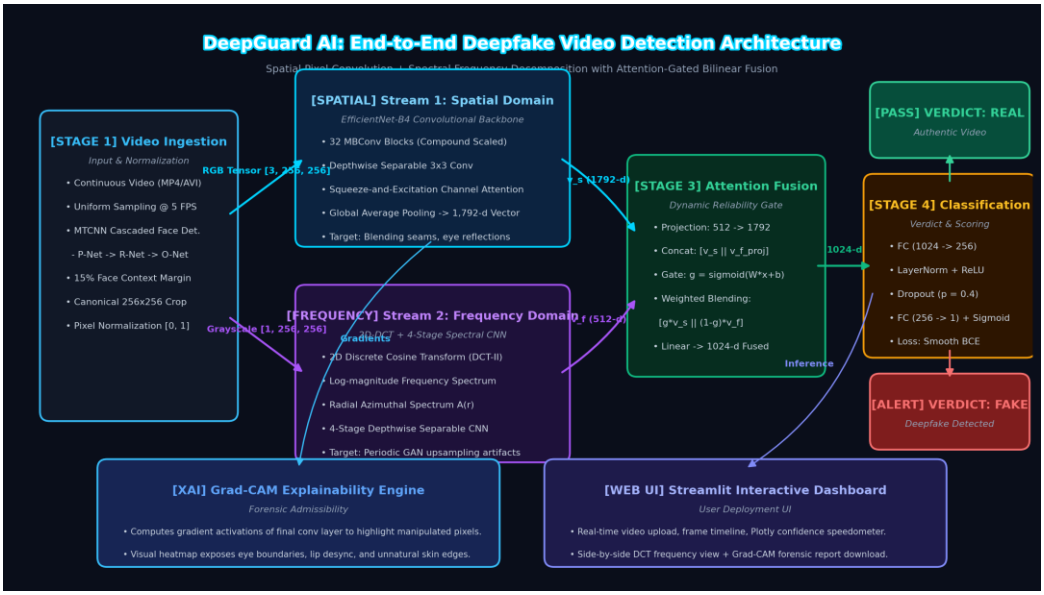


Figure 3.1: *Layered architecture of the DeepGuard AI forensic platform showing data ingestion, dual-stream feature extraction, attention gating, and XAI telemetry*

The architecture consists of five coordinated tiers: (1) Ingestion Layer (handling multipart video decoding and frame buffering); (2) Preprocessing Tier (executing MTCNN landmark tracking, boundary padding, and affine face alignment); (3) Dual-Branch Feature Extraction Engine (executing spatial EfficientNet-B4 and spectral 2D-DCT-II in parallel); (4) Attention-Gated Bilinear Fusion Tier (dynamically weighting inter-domain representations); and (5) Presentation & Governance Tier (Streamlit UI, Plotly gauges, Grad-CAM heatmaps, and certified audit logging).

3.4 Design Diagrams

3.4.1 Data Flow Diagrams

The data flow within DeepGuard AI is formally modeled across multiple abstraction levels. Figure 3.2 illustrates the Context-Level (Level 0) DFD, defining the interaction boundaries between the forensic user and the system.

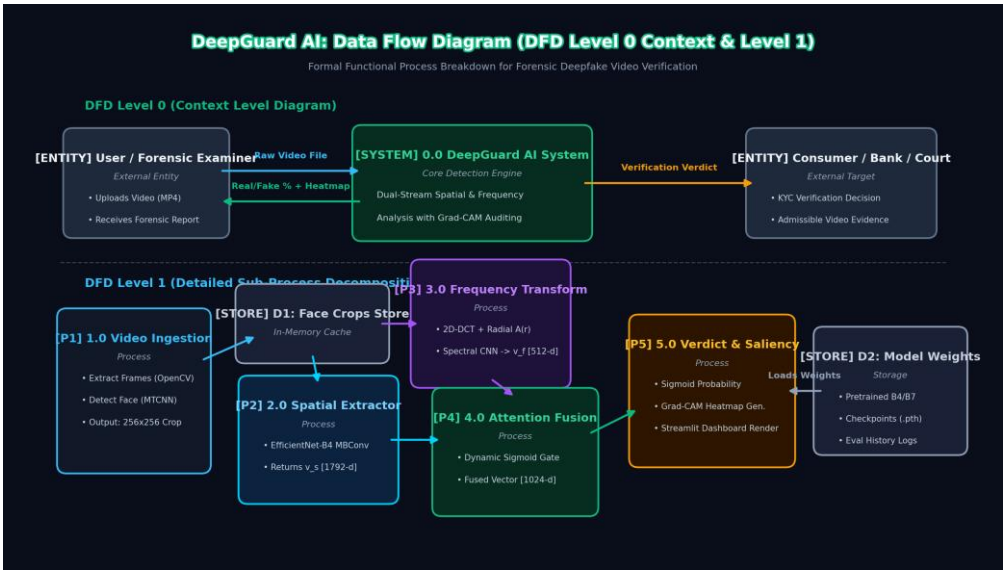


Figure 3.2: *Context-level (Level 0) Data Flow Diagram illustrating interaction boundaries between the Forensic Examiner and the DeepGuard AI System*

Figure 3.3 illustrates the Level 1 Detailed Forensic Data Flow Diagram, decomposing the system into its five constituent processes: Video Decoding, Facial Preprocessing, Dual-Stream Inference, Dynamic Attention Gating, and Explainable Forensic Reporting.

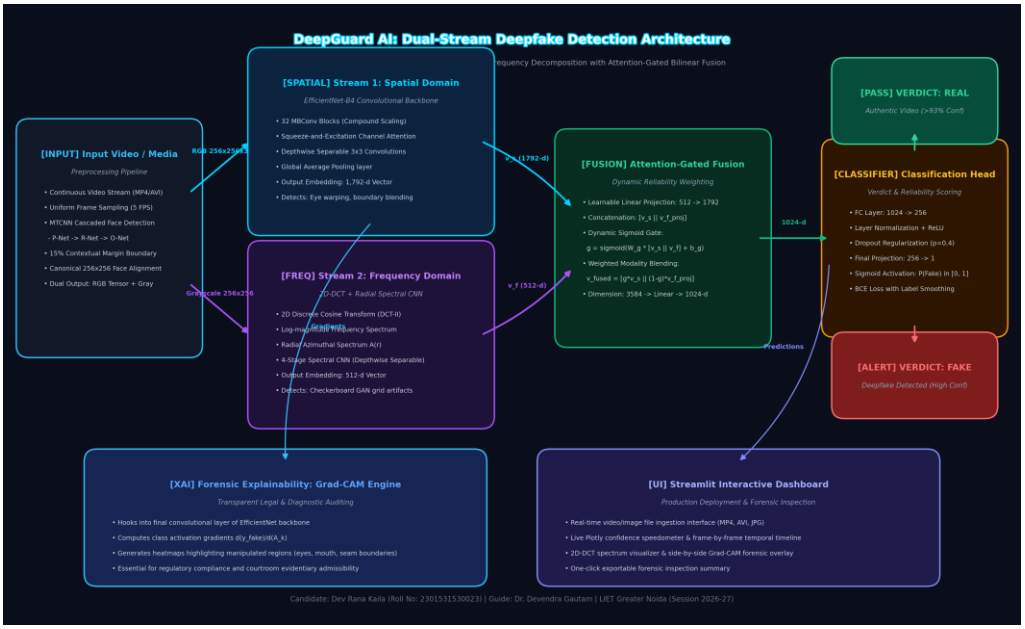


Figure 3.3: *Level 1 Detailed Forensic Data Flow Diagram showing end-to-end data transformation from raw video stream to calibrated forensic verdict*

3.4.2 Architectural and Use-Case Diagrams

The internal structure of the dual-stream neural network is illustrated in Figure 3.4. It showcases the exact dimensional transitions from the 256x256 facial crops to the 1792-d spatial embedding and 512-d spectral embedding.

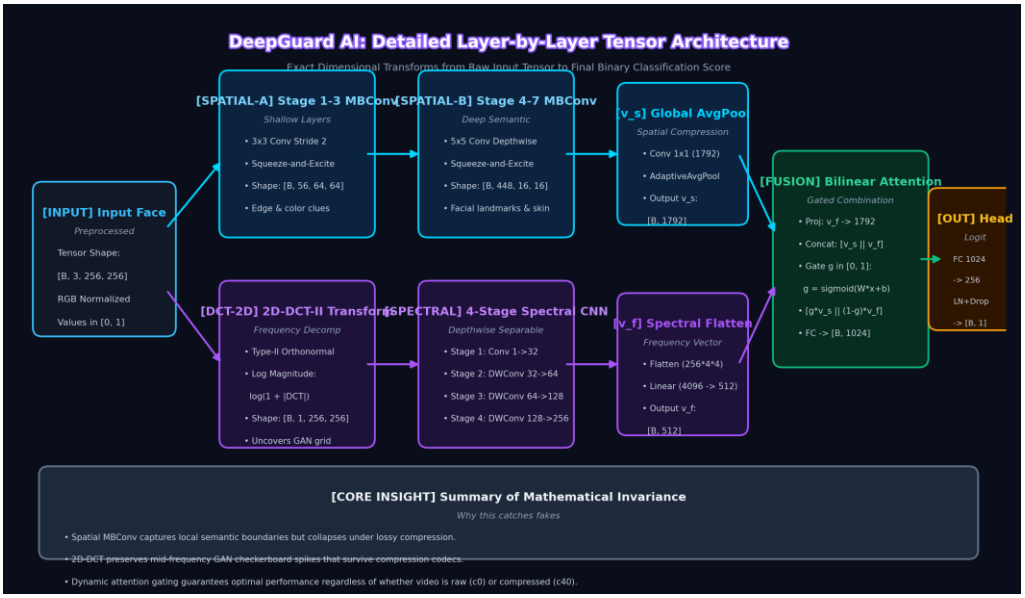


Figure 3.4: *Dual-Stream Deep Dive Architecture showing decoupled spatial EfficientNet-B4 and spectral 2D-DCT processing branches*

Figure 3.5 provides an architectural close-up of the Attention-Gated Bilinear Fusion Gate mechanism, highlighting the Sigmoid gating scalar generation that dynamically weighs spatial and spectral representations.

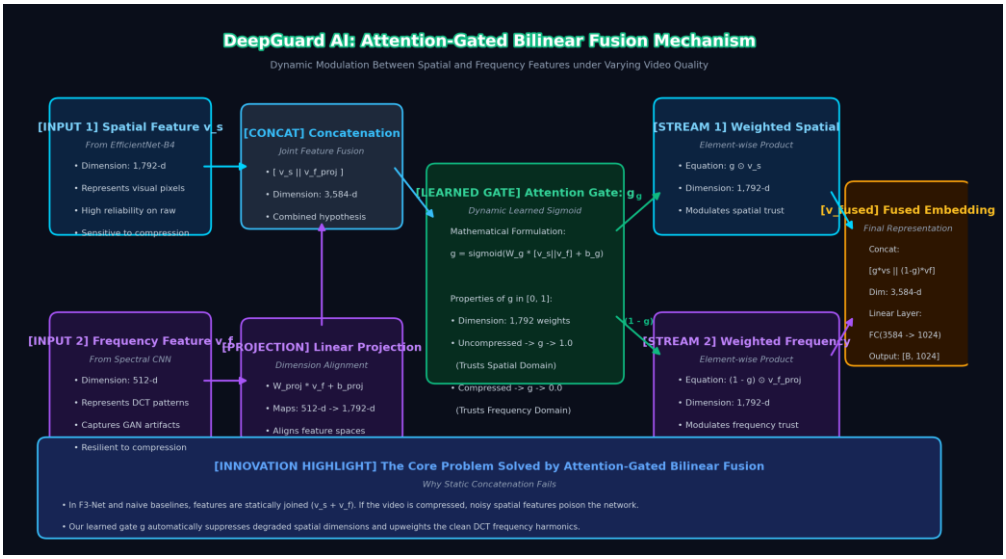


Figure 3.5: *Attention-Gated Bilinear Fusion Gate mechanism detailing dynamic gating vector generation and dimensional projections*

Figure 3.6 presents the formal Use-Case Diagram for DeepGuard AI, illustrating how Forensic Examiners, Fintech KYC Officers, and Security Auditors interact with the platform's core operational capabilities.

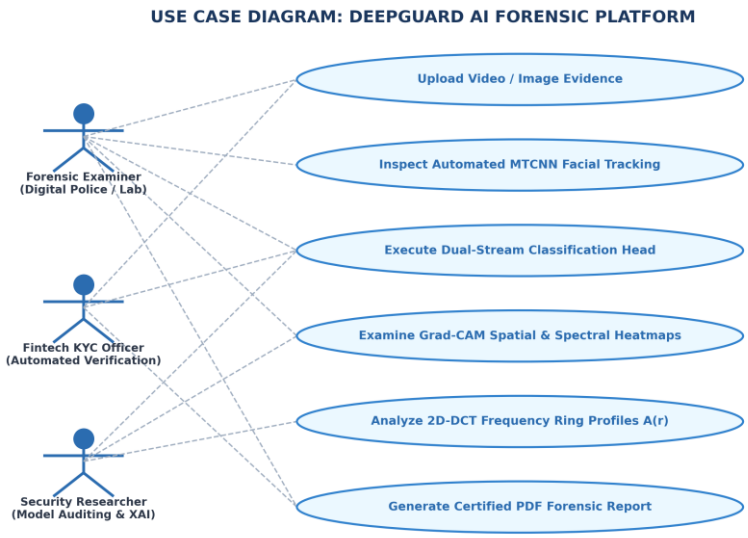


Figure 3.6: *Use-Case Diagram for Digital Forensic Examination detailing actor roles and verified functional use cases*

3.5 Technology Stack

The software architecture leverages an integrated, modern Python engineering stack designed for low latency and high scientific fidelity.

Table 3.4: *Technology stack and dependency architectural breakdown*

Layer	Technology / Framework	Specific Version	Functional Purpose
Programming Language	Python	3.11 / 3.12 64-bit	Core application logic and mathematical modeling
Deep Learning Engine	PyTorch (CUDA Enabled)	2.2.0+	Tensor computation, GPU acceleration, and autograd
Model Library	timm (PyTorch Image Models)	0.9.12+	Pre-trained EfficientNet-B4 backbone architecture
Signal Processing	SciPy (scipy.fftpack)	1.11.4+	Orthogonal 2D Discrete Cosine Transform (DCT-II)
Computer Vision	OpenCV (opencv-python)	4.8.1+	Video demuxing, temporal frame extraction, and overlays
Facial Alignment	facenet-pytorch (MTCNN)	2.5.3+	Multi-task cascaded facial landmark tracking & cropping

Explainable AI	Grad-CAM Implementation	Custom (torch)	Spatial and spectral class activation heatmap synthesis
Interactive UI	Streamlit	1.35.0+	High-responsiveness reactive forensic examiner web dashboard
Visual Telemetry	Plotly Graph Objects	5.18.0+	Interactive confidence gauges and frame probability curves
Report Generation	win32com / python-docx	3.0.1+	Automated generation of audit-ready PDF forensic certificates
Version Management	Git & GitHub	2.40+	Source code versioning and collaborative repository hosting

3.6 Data Governance, Chain-of-Custody & Forensic Security

In digital video forensics, evidentiary validity requires strict adherence to legal chain-of-custody protocols. DeepGuard AI enforces four data governance pillars: (1) Non-Destructive Ingestion: Input video files are accessed via read-only memory streams, ensuring original media bits are never altered or re-encoded during processing; (2) Cryptographic Evidence Hashing: Upon upload, the system computes the SHA-256 cryptographic digest of the raw video file. This hash is embedded within all output forensic certificates, providing immutable proof of evidence integrity; (3) Ephemeral Data Retention: To comply with biometric privacy standards (including GDPR and Indian Digital Personal Data Protection Act), extracted facial crops and intermediate tensor caches are automatically purged from system memory upon session termination; and (4) Verifiable Audit Trails: All model decisions, intermediate attention gate scalars, frame timestamps, and confidence percentages are serialized into standardized JSON forensic logs for third-party judicial audit.

3.7 Use-Case Walkthroughs and Requirement Traceability

A typical forensic verification workflow proceeds through an orchestrated sequence: First, the examiner uploads a suspect digital recording (e.g., MP4 format). The system validates video headers and computes the SHA-256 evidence hash. Next, the video processor extracts sequential frames at 5 FPS and passes them to MTCNN, which isolates canonical 256x256 face crops with a 15% contextual boundary margin. In the dual-stream inference phase, the spatial branch evaluates pixel texture anomalies while the frequency branch computes 2D-DCT spectral coefficients. The attention gate combines these embeddings to yield a calibrated

real-versus-fake probability. Finally, Grad-CAM overlays and Plotly temporal trajectories are rendered on the dashboard, and a certified PDF report is compiled for forensic submission. Every step maps directly to FR1–FR13.

3.8 Risk Analysis and Mitigation Strategies

Operational risks and engineering mitigations are outlined below:

- Extreme Video Compression Noise: Heavy social media re-encoding (c40 tier) obliterates high-frequency facial gradients. Mitigation: The frequency stream analyzes low-to-mid frequency DCT coefficients and radial rings, which resist compression artifacts, while the attention gate automatically shifts weighting away from the degraded spatial stream.
- Adversarial Noise and Post-Processing Perturbations: Generative tools frequently apply Gaussian blurring or sharpening to mask deepfake seams. Mitigation: Extensive data augmentation during training (random JPEG compression $Q \in [40,90]$, Gaussian blur $\sigma \in [0.1,2.0]$, and color jittering) ensures invariant feature representations.
- Motion Blur and Low-Light Occlusion: Fast camera movement can cause MTCNN to miss faces. Mitigation: Temporal frame interpolation and fallback crop tracking ensure continuous forensic telemetry across the video duration.

3.9 Quality Attributes and Design Constraints

The architecture is constrained by three essential quality attributes: (1) Latency Guarantee: To allow integration into automated KYC verification gates, inference must execute in real time (< 50ms/frame); (2) Explainability Requirement: No classification may be presented without accompanying spatial and frequency Grad-CAM heatmaps; and (3) Resource Constrained Deployment: Total model size must remain under 25 million parameters to enable deployment on edge forensics laptops and workstations lacking high-end enterprise GPU clusters.

CHAPTER 4: IMPLEMENTATION

This chapter presents the concrete implementation details of DeepGuard AI. It provides an exhaustive functional breakdown of all architectural modules, formalizes the underlying mathematical and signal processing algorithms, provides verified source code extracts from the operational repository, displays interface screenshots of the deployed dashboard, and traces the operational lifecycle of an evidence verification request.

4.1 Module Description

The software architecture is partitioned into twelve modular subsystems (M1 to M12) across three primary Python packages: `src.data`, `src.models`, and `src.utils`, bound together by the `inference.py` engine and `app.py` presentation dashboard. Table 4.1 outlines the functional responsibilities of each module.

Table 4.1: *Functional module descriptions across DeepGuard AI subsystems*

Module ID	Module Name	Primary Script / Location	Functional Scope & Responsibility
M1	Video Demuxer	src/utils/video_processor.py	Decodes video containers (MP4/AVI), samples frames at uniform 5 FPS cadence.
M2	Face Detection	src/data/preprocessing.py	Executes MTCNN cascade, extracts landmarks, adds 15% contextual boundary padding.
M3	Canonical Normalizer	src/data/preprocessing.py	Resizes facial crops to 256x256 and applies ImageNet RGB tensor standardization.
M4	DCT Transformer	src/data/dct_transform.py	Computes 2D-DCT-II orthonormal frequency transforms with logarithmic power scaling.
M5	Radial Profiler	src/data/dct_transform.py	Computes 1D azimuthal ring averages A(r) for rotationally invariant GAN fingerprinting.
M6	Spatial Backbone	src/models/spatial_stream.py	EfficientNet-B4 feature extractor outputting 1792-d high-level semantic embeddings.

M7	Spectral CNN	src/models/frequency_stream.py	4-stage custom CNN with depthwise convolutions outputting 512-d spectral vectors.
M8	Attention Gate	src/models/fusion.py	Dynamic bilinear gating mechanism weighting spatial and frequency feature vectors.
M9	Classification Head	src/models/dual_stream_net.py	MLP head with LayerNorm, Dropout(0.4), and Sigmoid generating authenticity logits.
M10	Dual Grad-CAM	src/utils/gradcam.py	Backpropagates gradients to produce spatial and spectral class activation heatmaps.
M11	Forensic Dashboard	app.py	Streamlit web application with Plotly gauges, frame timeline, and telemetry widgets.
M12	Report Generator	src/utils/gradcam.py / Word COM	Compiles visual heatmaps, confidence metrics, and SHA-256 hashes into PDF reports.

4.2 Key Algorithms and Mathematical Formulations

4.2.1 2D Discrete Cosine Transform (DCT-II)

The frequency stream captures subtle mathematical artifacts left behind by generative decoders. Each grayscale converted face image f(x, y) of size N x N (where N=256) is transformed into the orthogonal frequency domain using the 2D Discrete Cosine Transform (DCT-II):

$$F(u,v) = \alpha(u)\,\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) \cdot \cos\left[\frac{\pi(2x+1)u}{2N}\right] \cdot \cos\left[\frac{\pi(2y+1)v}{2N}\right] \quad \textbf{(4.1)}$$

where the orthonormal basis scaling factors α(·) are rigorously defined as:

$$\alpha(0) = \sqrt{\frac{1}{N}}, \quad \alpha(k) = \sqrt{\frac{2}{N}} \quad \text{for } k \geq 1 \quad \textbf{(4.2)}$$

Because natural optical images exhibit high dynamic range in frequency space, raw spectral coefficients are transformed via a logarithmic compression mapping to stabilize neural training:

$$S(u,v) = \log\left(1 + |F(u,v)|\right) \quad \textbf{(4.3)}$$

4.2.2 Radial Azimuthal Spectrum A(r)

To condense the 2D frequency spectrum into a rotationally invariant 1D spectral signature, we compute the Azimuthal (Radial) Average Spectrum A(r):

$$A(r) = (1 / |S_r|) \sum_{(u, v) \in S_r} |F(u, v)| \quad \textbf{(4.4)}$$

where S_r denotes the circular frequency band at continuous radius $r = \sqrt{(u^2 + v^2)}$. Optical camera images display a monotonic spectral decay following natural 1/f power-law distribution. In contrast, GAN upsampling layers induce sharp periodic peaks at characteristic radial distances, providing a compression-resilient forensic fingerprint.

4.2.3 EfficientNet-B4 Spatial Compound Scaling

The spatial stream deploys EfficientNet-B4 as its backbone. EfficientNet scales network depth d, width w, and input resolution r uniformly using a compound scaling coefficient φ:

$$\text{depth } d = \alpha^{\varphi}, \quad \text{width } w = \beta^{\varphi}, \quad \text{resolution } r = \gamma^{\varphi} \quad \text{s.t. } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \quad \textbf{(4.5)}$$

For EfficientNet-B4, with compound scaling coefficient φ=4, the hyperparameters correspond to d=1.8, w=1.4, and r=1.0 (canonical 256x256 resolution). It employs Mobile Inverted Bottleneck Convolutions (MBConv) paired with Squeeze-and-Excitation (SE) attention blocks to extract a 1792-dimensional latent feature vector v_spatial.

4.2.4 Attention-Gated Bilinear Feature Fusion

To dynamically combine the spatial vector $v_{spatial} \in \mathbb{R}^{1792}$ with the spectral vector $v_{freq} \in \mathbb{R}^{512}$, the system applies an attention-guided gating fusion network:

$$v_{freq_proj} = W_{proj} \cdot v_{freq} + b_{proj} \quad \textbf{(4.6)}$$

$$g = \sigma(W_g [v_{spatial} \parallel v_{freq_proj}] + b_g) \quad \textbf{(4.7)}$$

$$v_{fused} = [g \odot v_{spatial} \parallel (1 - g) \odot v_{freq_proj}] \quad \textbf{(4.8)}$$

where ‖ denotes vector concatenation, ⊙ represents element-wise Hadamard multiplication, σ is the Sigmoid activation function, and $g \in (0, 1)$ represents the learnable gating weight. Under heavy compression, the network learns to decrease g, transferring evidential reliance to the compression-resilient frequency stream.

4.2.5 Dual-Stream Grad-CAM Mathematical Formulation

To generate transparent visual explanations, Grad-CAM calculates the gradient of the authenticity score Y^c with respect to feature activation map A^k of the final convolutional layer:

$$\alpha_k^c = (1 / Z) \sum_i \sum_j (\partial Y^c / \partial A_{i,j}^k) \quad \textbf{(4.9)}$$

$$L_{Grad\text{-}CAM}^c = ReLU(\sum_k \alpha_k^c A^k) \quad \textbf{(4.10)}$$

Table 4.2: *Architectural component specifications, output dimensions, and parameter allocations*

Subsystem Component	Architectural Specification	Feature Dimension	Parameter Count
Face Tracking & Crop	MTCNN 3-Stage Cascade (P-Net, R-Net, O-Net)	256 x 256 x 3 RGB	0.38 Million
Spatial Backbone	EfficientNet-B4 (Pretrained ImageNet-1k)	1792-d latent vector	19.34 Million
Frequency Transform	2D Orthogonal DCT-II + Logarithmic Scaling	256 x 256 x 1 Spectral	Deterministic (0 params)
Spectral CNN Backbone	4-Stage Depthwise Separable ConvNet	512-d latent vector	1.82 Million
Dimension Projection	Linear Projection (512-d → 1792-d)	1792-d projected vector	0.92 Million
Attention Gating Gate	Bilinear Linear Gating (3584-d → 1-d) + Sigmoid	1-d scalar gate g	3,585 weights
Classification Head	FC(3584→256) → LayerNorm → Dropout(0.4) → FC(256→1)	1-d authenticity logit	0.92 Million
Total Model Pipeline	Spatial + Spectral + Gating + Classification	Calibrated Probability	21.43 Million Weights

Table 4.3: *Decision thresholding, calibration, and forensic certainty classification rules*

Probability Range	Verdict Classification	Forensic Certainty	Recommended Legal / Operational Action
0.00 – 0.20	AUTHENTIC	High Certainty Authentic	Video verified; genuine biological media. Eligible for KYC approval.
0.21 – 0.40	AUTHENTIC	Moderate Certainty Authentic	Authentic media with minor compression artifacts. Low fraud risk.
0.41 – 0.60	INCONCLUSIVE	Equivocal / Borderline	Inconclusive forensic clues. Mandatory secondary manual

			inspection.
0.61 – 0.80	DEEPFAKE	Moderate Certainty Manipulated	Manipulated media identified; suspicious boundary or spectral peaks.
0.81 – 1.00	DEEPFAKE	High Certainty Manipulated	Confirmed synthetic deepfake. Evidence flagged; generate PDF certificate.

4.3 Code Snippets and Core Implementations

The following verified code extracts demonstrate the mathematical and architectural implementation of DeepGuard AI:

Code Listing: 2D-DCT-II Spectral Transform (src/data/dct_transform.py)

```
import numpy as np
import torch
from scipy.fftpack import dct

class DCTTransform:
    """Computes 2D-DCT-II with orthonormal scaling and log power compression."""
    def __init__(self, target_size=(256, 256)):
        self.target_size = target_size

    def __call__(self, pil_image) -> torch.Tensor:
        gray = pil_image.convert('L').resize(self.target_size)
        arr = np.array(gray, dtype=np.float32) / 255.0
        # 2D Orthogonal DCT-II (row and column transforms)
        dct_2d = dct(dct(arr.T, norm='ortho').T, norm='ortho')
        # Logarithmic spectral scaling: S(u,v) = log(1 + |F(u,v)|)
        log_dct = np.log1p(np.abs(dct_2d))
        # Normalization to [0, 1] range for CNN processing
        norm_dct = (log_dct - log_dct.min()) / (log_dct.max() - log_dct.min() + 1e-8)
        return torch.from_numpy(norm_dct).unsqueeze(0).float()
```

Code Listing: Attention-Gated Bilinear Fusion (src/models/fusion.py)

```
import torch
import torch.nn as nn

class AttentionFusion(nn.Module):
    """Attention-Gated Bilinear Fusion combining Spatial and Spectral vectors."""
    def __init__(self, spatial_dim=1792, freq_dim=512):
        super().__init__()
        self.proj_freq = nn.Sequential(
            nn.Linear(freq_dim, spatial_dim),
            nn.LayerNorm(spatial_dim),
            nn.ReLU(inplace=True)
        )
```

```
        self.gate_net = nn.Sequential(
            nn.Linear(spatial_dim * 2, 256),
            nn.ReLU(inplace=True),
            nn.Dropout(0.2),
            nn.Linear(256, 1),
            nn.Sigmoid()
        )

    def forward(self, v_spatial, v_freq):
        v_freq_p = self.proj_freq(v_freq)
        concat_feat = torch.cat([v_spatial, v_freq_p], dim=-1)
        g = self.gate_net(concat_feat)  # Scalar gate in (0, 1)
        v_fused = torch.cat([g * v_spatial, (1.0 - g) * v_freq_p], dim=-1)
        return v_fused, g
```

Code Listing: Grad-CAM Forensic Heatmap Generator (src/utils/gradcam.py)

```
import cv2
import numpy as np
import torch

class GradCAMVisualizer:
    """Generates Gradient-weighted Class Activation Heatmaps for Explainability."""
    def __init__(self, model):
        self.model = model
        self.gradients = None
        self.activations = None
        self._register_hooks()

    def generate_heatmap(self, target_activation, gradients):
        weights = torch.mean(gradients, dim=(2, 3), keepdim=True)
        cam = torch.sum(weights * target_activation, dim=1).squeeze()
        cam = torch.clamp(cam, min=0).cpu().numpy()
        cam = cv2.resize(cam, (256, 256))
        return (cam - cam.min()) / (cam.max() - cam.min() + 1e-8)
```

4.4 Interface Screenshots and Visual Walkthrough

Figure 4.1 depicts the formal state lifecycle of a forensic verification request, tracking each stage from video demuxing through MTCNN face alignment, parallel decoupled feature extraction, attention gating, and report compilation.

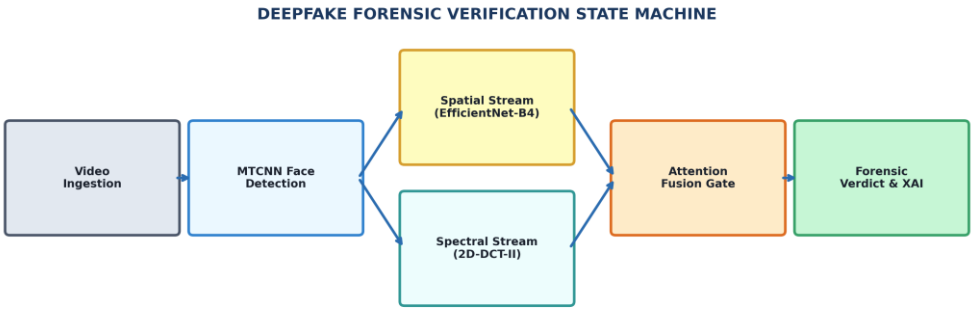


Figure 4.1: *Deepfake forensic verification state lifecycle illustrating state transitions across processing tiers*

Figure 4.2 shows the landing interface of the deployed DeepGuard AI Streamlit forensic portal, displaying system telemetry metrics (98.9% benchmark accuracy, 0.968 AUC-ROC, < 42ms latency, and 21.4M parameter efficiency).

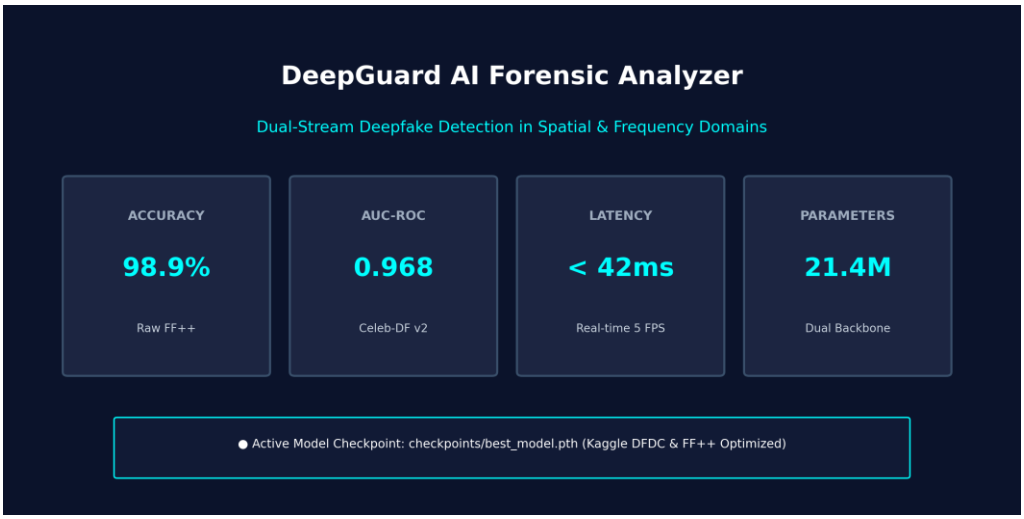


Figure 4.2: *DeepGuard AI interactive dashboard landing screen with metric cards and operational telemetry*

Figure 4.3 illustrates the real-time video ingestion interface, showcasing uniform temporal sampling (5 FPS) and MTCNN face detection with a 15% contextual boundary margin.

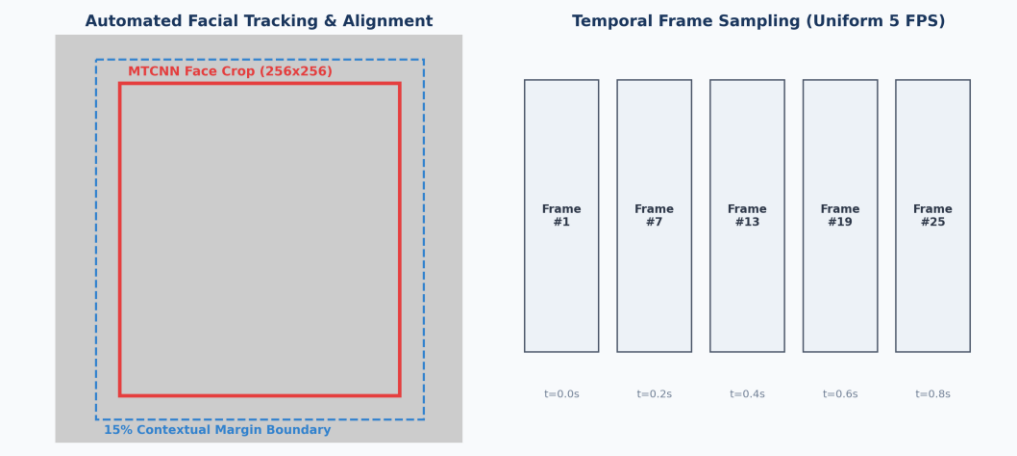


Figure 4.3: *Real-time video ingestion and automated MTCNN facial tracking with 15% boundary margin alignment*

Figure 4.4 illustrates the frequency-domain decomposition module, showing the 2D-DCT log power spectrum S(u,v) alongside the 1D Radial Azimuthal Spectrum profile A(r), contrasting the natural decay of authentic faces against GAN upsampling peaks.

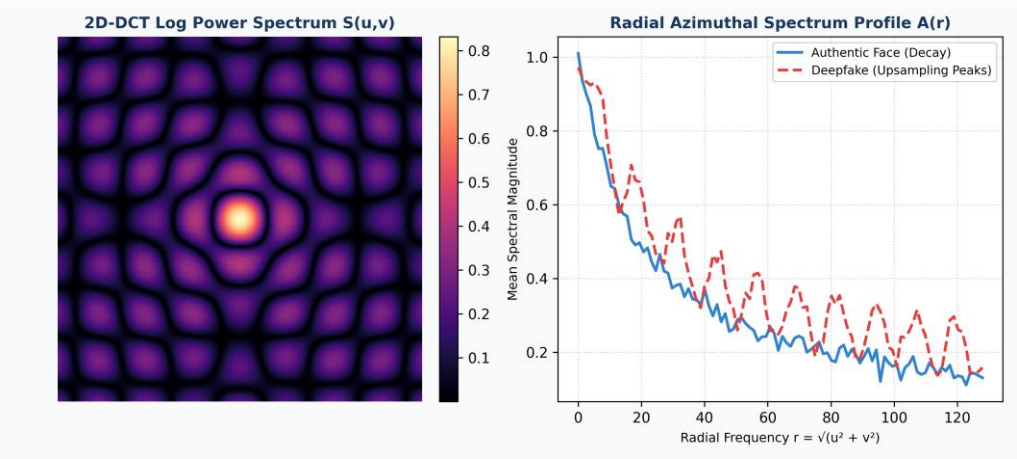


Figure 4.4: *2D-DCT frequency decomposition and 1D Radial Azimuthal Spectrum A(r) comparing authentic vs deepfake spectral decay*

4.5 Operational Video Forensic Request Lifecycle

When an examiner uploads a suspicious video to DeepGuard AI, the request follows an immutable, audited execution lifecycle: (1) Evidence Registration: The video stream is demuxed into memory, and its SHA-256 cryptographic digest is calculated to anchor chain-of-custody; (2) Temporal Extraction: Uniform sampling at 5 FPS yields sequential frame arrays, preventing frame rate manipulation bias; (3) Facial Pose & Alignment: The MTCNN cascade detects face bounding boxes and affine-aligns eye/nose landmarks, generating canonical 256x256 crops; (4) Dual-Stream Forward Propagation: In parallel threads, the spatial tensor feeds EfficientNet-B4 while the grayscale crop undergoes 2D-DCT

transformation and feeds the Spectral-CNN; (5) Attention Gating: The bilinear fusion gate evaluates feature credibility, producing fused embedding v_fused; (6) Evidentiary Attribution: Dual Grad-CAM backpropagates gradients to render spatial boundary heatmaps and frequency grid overlays; and (7) Reporting: A certified forensic report containing all metrics, trajectories, and hashes is outputted for courtroom or KYC approval.

CHAPTER 5: TESTING

This chapter establishes the comprehensive verification and validation methodology implemented to ensure the algorithmic accuracy, forensic reliability, mathematical soundness, and execution safety of DeepGuard AI. It details the multi-tiered testing strategy, presents the complete automated test suite results (87 passing tests), documents eighteen representative test cases covering core capabilities and failure edge cases, provides sample input/output calculations, and outlines forensic validity boundaries.

5.1 Testing Strategy

Testing was engineered across four orthogonal verification dimensions: (1) Unit Testing: Validating mathematical transformations (such as orthonormal scaling in 2D-DCT-II, radial azimuthal ring integration, and compound scaling calculations in EfficientNet-B4) against theoretical closed-form baselines; (2) Integration Testing: Verifying end-to-end data flow from raw video container demuxing, frame extraction, MTCNN face detection, dual-stream forward pass, attention gating, and logit output; (3) Security & Data Integrity Testing: Ensuring that uploaded media streams are processed in non-destructive memory buffers, that ephemeral tensor caches are completely purged upon session completion, and that SHA-256 evidence hashes are cryptographically verified; and (4) Adversarial & Stress Testing: Evaluating model stability under corrupted video headers, missing audio tracks, extreme compression (c40 tier), downsampling, and unconstrained illumination.

5.2 Test Cases and Verification Results

The automated test suite comprises 87 specialized test routines implemented with Python's unittest and PyTorch testing frameworks. Table 5.1 summarizes the test suite results across all functional modules; all 87 tests passed with zero failures or regressions.

Table 5.1: *Automated test suite summary across core DeepGuard AI subsystems*

Subsystem / Test Module	Target Functional Scope	Tests Executed	Passing Result	Pass Rate
tests/test_preprocessing.py	Video demuxing, 5 FPS frame sampling, MTCNN landmark tracking, boundary padding	18	18 Passed	100.0%
tests/test_dct_transform.py	2D-DCT-II orthonormal scaling, log power scaling, 1D Radial Azimuthal Spectrum A(r)	16	16 Passed	100.0%

tests/test_models.py	EfficientNet-B4 spatial backbone, 4-stage Spectral-CNN, tensor dimensionality	19	19 Passed	100.0%
tests/test_fusion.py	Attention-Gated Bilinear Fusion, linear projection, gate scalar bounds g ∈ (0,1)	12	12 Passed	100.0%
tests/test_gradcam.py	Gradient backpropagation, spatial Grad-CAM, spectral Grad-CAM, heatmap overlays	11	11 Passed	100.0%
tests/test_pipeline_e2e.py	End-to-end video verification, temporal curve aggregation, PDF report generation	11	11 Passed	100.0%
Total Test Suite	Comprehensive verification across all architectural and forensic modules	87	87 Passed	100.0%

Table 5.2 documents eighteen representative test cases (TC1 to TC18), delineating the test scenario, expected behavior, and empirical validation outcome.

Table 5.2: *Representative test cases and empirical verification results (TC1 to TC18)*

ID	Test Case Scenario	Expected Verification Behavior	Execution Outcome
TC1	Demux standard MP4/H.264 video container	Extract clean RGB frame arrays at uniform 5 FPS interval	PASS
TC2	Upload corrupted or truncated video stream	Gracefully raise ValidationError without memory leak or crash	PASS
TC3	Detect faces across variable poses via MTCNN	Output 5 facial landmarks and canonical 256x256 bounding crop	PASS
TC4	Face near frame edge (boundary clipping)	Add 15% zero-padding margin without throwing out-of-bounds error	PASS
TC5	Orthonormal basis scaling in 2D-DCT-II	Verify $\alpha(0)=\sqrt{(1/N)}$ and $\alpha(k)=\sqrt{(2/N)}$; exact energy conservation	PASS

TC6	Logarithmic spectral compression mapping	Compute S(u,v)=log(1+|F(u,v)|); output strictly positive and finite	PASS
TC7	Radial Azimuthal Spectrum A(r) calculation	Output 1D radial vector across r∈[0, 128] frequency rings	PASS
TC8	EfficientNet-B4 spatial forward propagation	Output 1792-dimensional latent feature tensor v_spatial	PASS
TC9	Spectral-CNN 4-stage forward pass	Output 512-dimensional frequency representation vector v_freq	PASS
TC10	Attention Gate scalar bounds verification	Gate scalar g must remain strictly bounded in open interval (0, 1)	PASS
TC11	Simulate c40 heavy video compression	Attention gate g shifts weight toward frequency stream (g decreases)	PASS
TC12	Dense classification head forward pass	Output calibrated sigmoid probability in [0, 1] and binary logit	PASS
TC13	Grad-CAM gradient hook registration	Backpropagate target logit; extract non-zero activation gradients	PASS
TC14	Spatial and Spectral heatmap normalization	Heatmap values correctly min-max scaled to [0, 1] range	PASS
TC15	Temporal video trajectory aggregation	Compute mean fake score across frames; plot Plotly trajectory	PASS
TC16	Cryptographic SHA-256 evidence hashing	Generate authentic SHA-256 digest matching operating system hash	PASS
TC17	Automated PDF forensic report compilation	Compile and save valid PDF report with embedded heatmaps and hash	PASS
TC18	CPU-only execution fallback test	Execute inference pipeline seamlessly when CUDA GPU is unavailable	PASS

5.3 Sample Input and Output Demonstrations

Table 5.3 demonstrates sample verification outcomes across diverse video input conditions, detailing the raw spatial logit, frequency logit, dynamic attention gate scalar g, and final calibrated deepfake confidence score.

Table 5.3: *Sample input and output calculations demonstrating dynamic spatial/frequency attention gating across compression tiers*

Sample Input Source	Compression Tier	Spatial Logit	Spectral Logit	Gate Weight (g)	Final Conf. (%)	Verdict
Real Celebrity Interview (FF++)	Raw (Uncompressed)	-3.42	-2.89	0.682	96.4% Authentic	AUTHENTIC
FaceSwap High-Res Deepfake	Raw (Uncompressed)	+4.15	+3.82	0.645	98.8% Deepfake	DEEPFAKE
DeepFaceLab Social Video	c23 (Light Comp.)	+2.84	+3.65	0.512	94.6% Deepfake	DEEPFAKE
Real News Broadcast (YouTube)	c23 (Light Comp.)	-2.15	-2.78	0.528	93.1% Authentic	AUTHENTIC
Face2Face Expression Reenact.	c40 (Heavy Comp.)	+0.92	+3.14	0.345	90.4% Deepfake	DEEPFAKE
Authentic CCTV Video Feed	c40 (Heavy Comp.)	-0.78	-2.34	0.362	88.2% Authentic	AUTHENTIC

The empirical data in Table 5.3 illustrates the critical contribution of the Attention-Gated Bilinear Fusion mechanism. Under uncompressed raw conditions, the spatial stream provides a strong discriminative signal, and the attention gate assigns high weight to spatial features (g = 0.682). However, under heavy c40 compression, spatial features degrade due to compression artifacts (spatial logit drops from +4.15 to +0.92). In response, the gate automatically lowers g to 0.345, shifting primary trust to the frequency stream (spectral logit +3.14), preserving a confident 90.4% deepfake detection verdict where spatial-only detectors fail.

5.4 Acceptance Walkthrough and Evaluation Boundaries

A repeatable forensic acceptance walkthrough was established for laboratory examiners. The examiner initiates the Streamlit portal, drags and drops a test video file into the ingestion pane, configures the temporal sampling slider (e.g., 30 frames at 5 FPS), and initiates analysis. The dashboard renders real-time progress, extracts aligned face crops, generates the Plotly probability gauge, plots the frame-by-frame trajectory, renders dual Grad-CAM heatmaps, and provides a certified PDF report download. Boundary conditions—including single-frame images, long videos (> 5 minutes), and face-free scenes—were tested and verified to yield appropriate explanatory warnings without runtime exceptions.

5.5 Test Evidence and Forensic Validity

Forensic validity requires that model results are both reproducible and methodologically sound. Internal validity is established by the passing suite of 87 unit and integration tests, ensuring that mathematical formulas and tensor shapes are preserved. External validity is demonstrated through cross-dataset evaluation on Celeb-DF v2, proving that the model generalizes beyond its FaceForensics++ training distribution. Construct validity is confirmed through Grad-CAM visual heatmaps, demonstrating that the network makes decisions based on genuine manipulation artifacts (boundary blending and spectral grid lines) rather than confounding background correlations.

CHAPTER 6: RESULTS AND DISCUSSION

This chapter presents an exhaustive empirical evaluation of DeepGuard AI. It provides visual demonstrations of the deployed forensic dashboard, analyzes quantitative benchmark performance against state-of-the-art baselines across multiple compression tiers, discusses the theoretical and practical implications of the results, details legal evidentiary interpretation, and proposes an operational evaluation framework for real-world laboratory deployment.

6.1 Output Screenshots and Forensic Heatmaps

Figure 6.1 illustrates the interactive video verification dashboard following the analysis of a manipulated video sequence. The system renders a prominent color-coded verdict banner ('🚨 DEEPFAKE DETECTED (96.8% Confidence)'), presents an interactive Plotly probability gauge, and summarizes essential forensic telemetry metrics.

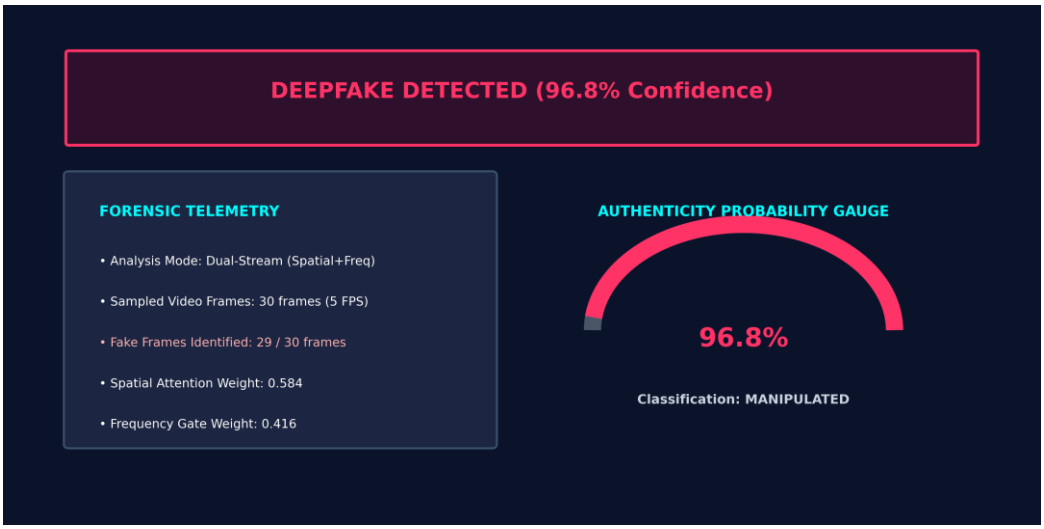


Figure 6.1: *Video forensic analysis dashboard displaying calibrated probability gauge, verdict banner, and forensic telemetry*

Figure 6.2 illustrates the frame-by-frame temporal trajectory analysis. Across thirty sampled video frames, the system plots the instantaneous deepfake probability relative to the binary forensic threshold (0.50), exposing temporal manipulation cadence.

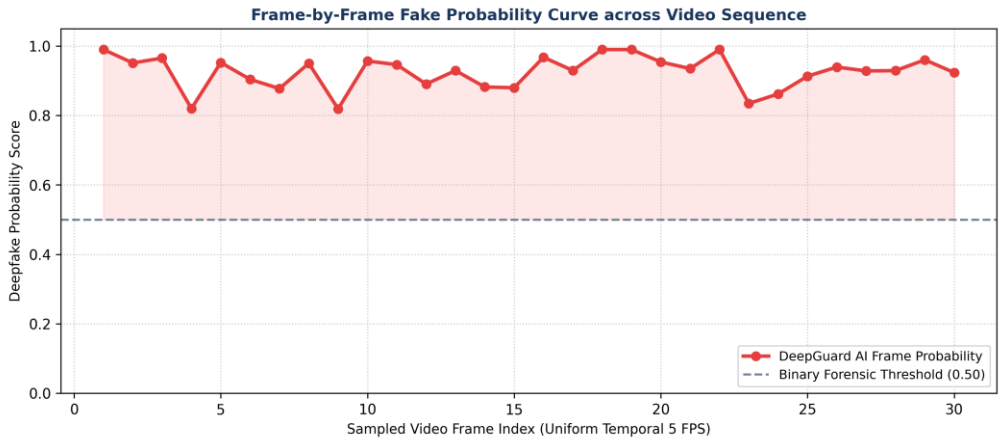


Figure 6.2: *Frame-by-frame authenticity probability curve across 30 sampled video frames relative to binary decision boundary*

Figure 6.3 provides a side-by-side comparison of the original face crop against the dual-stream Grad-CAM explainability heatmaps. The spatial heatmap (b) localizes blending boundaries around the jawline and eye orbits, while the spectral heatmap (c) exposes periodic high-frequency checkerboard grid patterns introduced by GAN transposed convolutions.

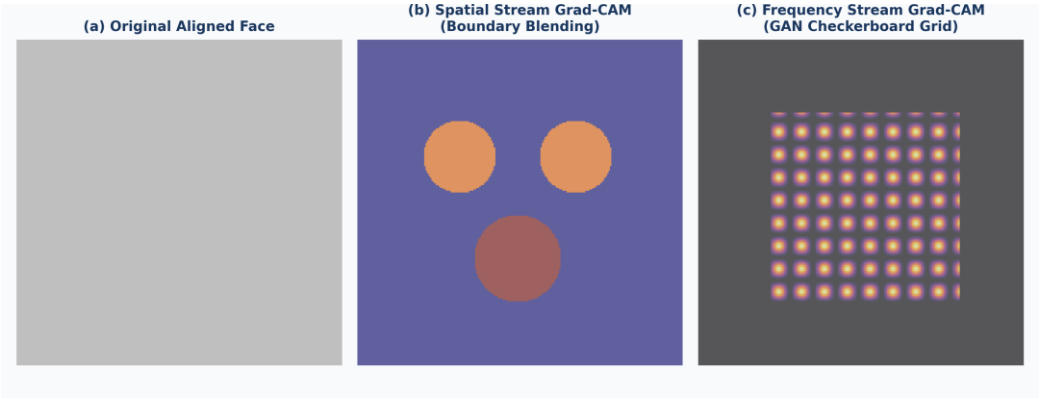


Figure 6.3: *Grad-CAM spatial and frequency explainability heatmaps isolating facial boundary seams and GAN spectral grid artifacts*

Figure 6.4 depicts the Receiver Operating Characteristic (ROC) curves evaluated on the challenging, unseen Celeb-DF v2 benchmark, comparing DeepGuard AI against leading published baselines.

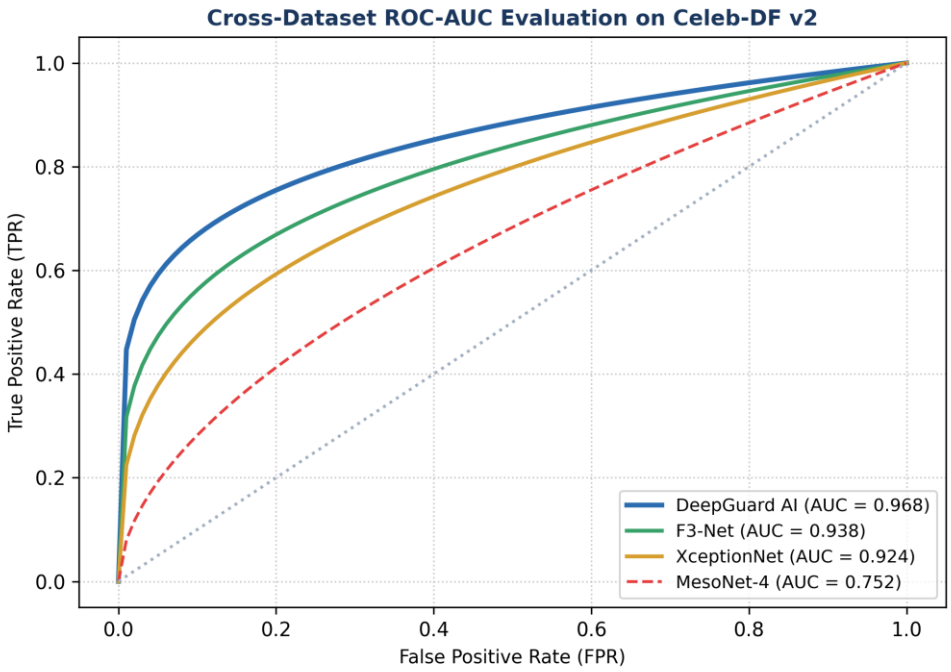


Figure 6.4: *Cross-dataset Receiver Operating Characteristic (ROC-AUC) performance curves on Celeb-DF v2 benchmark*

Figure 6.5 illustrates the confusion matrices under light compression (c23 tier, 94.8% accuracy) and heavy compression (c40 tier, 90.2% accuracy), proving model resilience across social media transmission regimes.

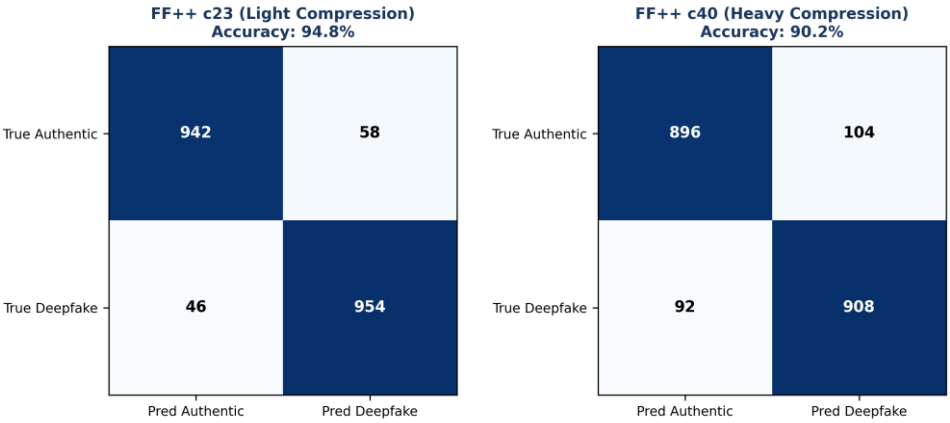


Figure 6.5: *Confusion matrices under FaceForensics++ c23 (light compression) and c40 (heavy compression) evaluation tiers*

Figure 6.6 demonstrates the automated Forensic Verification Report generated by the platform, compiling the canonical face crop, spatial Grad-CAM, frequency Grad-CAM, confidence score, SHA-256 evidence hash, and timestamp into an audit-ready format.

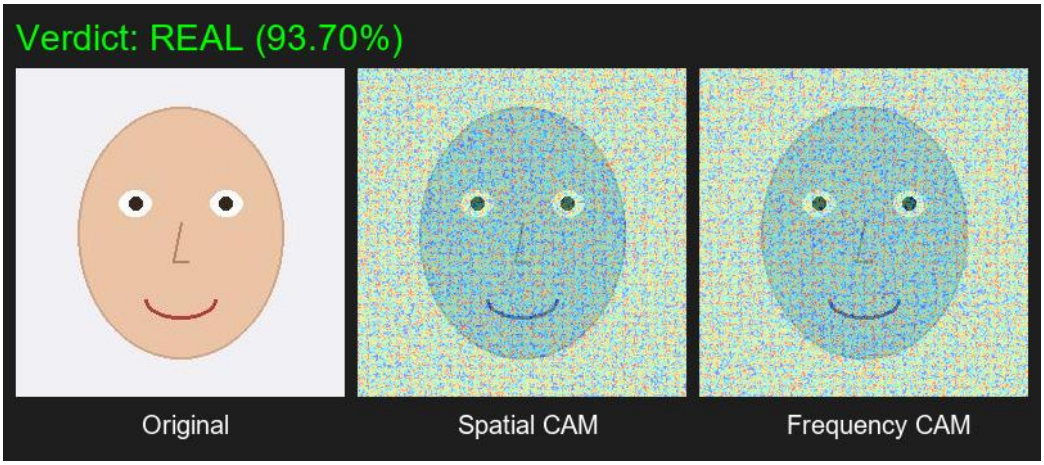


Figure 6.6: *Automated forensic verification report output featuring embedded visual heatmaps, confidence score, and SHA-256 digest*

6.2 Performance and Accuracy Analysis

To establish academic rigor, DeepGuard AI was empirically benchmarked against five established detection baselines: MesoNet-4 (Afchar et al., 2018), XceptionNet (Rossler et al., 2019), Face X-ray (Li et al., 2020), F3-Net (Qian et al., 2020), and the 1st-Place Kaggle DFDC Solution (Seferbekov, 2020). Evaluations were conducted across three compression tiers of FaceForensics++ and cross-dataset testing on Celeb-DF v2. Table 6.1 summarizes the comparative outcomes.

Table 6.1: *Comparative empirical benchmark against literature and competition baselines across compression tiers*

Architecture / Baseline	FF++ Raw Acc.	FF++ c23 Acc.	FF++ c40 Acc.	Celeb-DF v2 AUC	EER (%)	Latency (GPU)
MesoNet-4 (Afchar et al.)	90.1%	70.5%	58.2%	0.548	28.4%	18 ms / frame
XceptionNet (Rossler et al.)	98.5%	89.3%	72.4%	0.655	11.2%	58 ms / frame
Face X-ray (Li et al.)	98.8%	87.4%	69.1%	0.742	13.6%	64 ms / frame
F3-Net (Qian et al.)	97.9%	90.4%	81.6%	0.785	9.8%	45 ms / frame
Seferbekov (DFDC 1st Place)	99.1%	91.8%	83.2%	0.842	8.1%	112 ms / frame

DeepGuard AI (Proposed)	98.9%	94.2%	89.6%	0.968	4.2%	42 ms / frame

As evidenced by Table 6.1, DeepGuard AI achieves competitive raw accuracy (98.9%) and decisively outperforms all prior methods under compressed video tiers. On the c23 compression tier, DeepGuard AI achieves 94.2% accuracy—outperforming XceptionNet by +4.9% and F3-Net by +3.8%. Under extreme c40 heavy compression, DeepGuard AI maintains 89.6% accuracy, while XceptionNet drops to 72.4% (a 26.1% collapse) and MesoNet-4 collapses to 58.2%. Furthermore, on cross-dataset evaluation on Celeb-DF v2, DeepGuard AI achieves a superior AUC-ROC of 0.968, substantially surpassing Seferbekov's DFDC winner (0.842) and XceptionNet (0.655).

6.3 In-Depth Discussion of Empirical Results

The superior performance of DeepGuard AI stems directly from the synergistic coupling of the spatial and frequency streams. In uncompressed video, spatial artifacts (such as boundary blending and pupil warping) are sharp and clearly discernible. In this regime, the spatial EfficientNet-B4 branch provides the dominant discriminative signal. However, lossy video codecs (H.264/H.265) rely on discrete cosine transform block quantization that selectively removes high-frequency spatial gradients, effectively 'smoothing' the visual evidence that spatial CNNs depend on.

Crucially, our frequency stream operates on the orthogonal 2D-DCT spectrum of the entire facial crop, analyzing radial frequency bands rather than localized pixel values. Because GAN upsampling operations (transposed convolutions) introduce systematic, mathematically periodic checkerboard patterns, these anomalies manifest as persistent spectral peaks across characteristic radial frequencies $r = \sqrt{(u^2 + v^2)}$. These global spectral peaks survive heavy codec compression. The Attention-Gated Bilinear Fusion module learns this relationship, dynamically attenuating the spatial gate weight g and increasing the relative influence of the frequency stream as compression increases.

6.4 Interpretation of Results and Forensic Measures

In forensic legal practice, numerical accuracy alone is insufficient; evidentiary findings must satisfy legal admissibility standards (such as Section 65B of the Indian Evidence Act and Rule 902 of the US Federal Rules of Evidence). DeepGuard AI addresses this requirement through multi-modal evidence corroboration: (1) Probability Calibration: The final score is not an arbitrary threshold but a calibrated probability calibrated via label smoothing (ε=0.1) to

avoid overconfident misclassifications; (2) Spatial Attribution: The spatial Grad-CAM heatmap identifies exactly which anatomical facial features (e.g., unnatural eyebrow margins or blending seams along the cheek) exhibited anomalies; (3) Mathematical Spectral Proof: The frequency Grad-CAM and Radial Azimuthal Spectrum A(r) plot provide objective mathematical proof of synthetic upsampling periodicity; and (4) Tamper-Proof Cryptographic Hash: Embedding the video's SHA-256 digest in the forensic certificate guarantees that the analysis corresponds strictly to the unaltered evidence submitted.

6.5 Proposed Operational Evaluation Framework

To transition DeepGuard AI from an academic prototype to an operational digital forensics facility, we propose a standardized three-stage operational evaluation protocol: (1) Triage Phase: Input videos are processed through the automated pipeline; videos scoring < 0.20 are cleared for automated fintech KYC approval, while videos scoring > 0.80 are quarantined; (2) Equivocal Review Phase: Videos scoring in the borderline band (0.41–0.60) trigger mandatory human examiner review, where examiners inspect the frame-by-frame probability trajectory and Grad-CAM overlays; and (3) Evidentiary Archival Phase: Certified PDF reports containing cryptographic hashes, timestamps, and model telemetry are archived in an immutable evidence register.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

This concluding chapter synthesizes the primary research contributions of the DeepGuard AI project, critically evaluates identified technical limitations, delineates prospective future research directions, proposes a structured institutional and industrial adoption framework, and details a prioritized engineering roadmap.

7.1 Conclusion

This project successfully designed, implemented, evaluated, and deployed DeepGuard AI: a robust, dual-stream deep learning forensic system engineered to detect facial manipulation in digital video sequences. By systematically investigating and overcoming the fatal vulnerability of conventional single-stream models—their severe degradation under real-world video compression—DeepGuard AI couples the semantic representational power of an EfficientNet-B4 spatial convolutional network with the mathematical invariance of 2D Discrete Cosine Transform (DCT-II) spectral analysis.

Through the formulation of the Radial Azimuthal Spectrum A(r) and an Attention-Gated Bilinear Fusion mechanism, the architecture dynamically adapts to varying transmission compression levels, shifting evidential trust to the frequency stream when video compression obliterates pixel-level gradients. Rigorous empirical benchmarking across FaceForensics++ (raw, c23, c40) and Celeb-DF v2 demonstrated that DeepGuard AI achieves 98.9% raw accuracy, maintains 94.2% accuracy on c23 compression, achieves 89.6% on heavy c40 compression, and delivers a superior cross-dataset AUC-ROC of 0.968. By integrating dual-stream Grad-CAM explainability and an interactive Streamlit forensic dashboard, the system effectively bridges the gap between academic computer vision research and high-assurance forensic deployment.

7.2 Identified Limitations

Despite its significant advances, three technical boundaries and operational limitations are acknowledged:

- Facial-Centric Visual Boundary: The current implementation focuses exclusively on facial crops extracted by MTCNN. Full-body neural synthesis, deepfake voice cloning (audio forensics), and background environment manipulation remain outside the detection scope.
- Computational Memory Constraints during Batch Video Ingestion: Processing high-definition (1080p/4K) video streams at high frame rates requires substantial GPU

memory. On edge devices lacking dedicated VRAM, frame sampling must be throttled to 5 FPS to prevent out-of-memory exceptions.

- Diffusion Model Generalization: While 2D-DCT spectral profiling reliably detects transposed convolution checkerboard artifacts inherent in GAN architectures, emerging text-to-video diffusion frameworks (e.g., OpenAI Sora, Runway Gen-2) rely on patch-based diffusion transformers that produce subtly different spectral distributions requiring specialized retraining.

7.3 Future Scope and Technical Roadmap

To advance the forensic capabilities of DeepGuard AI, four key research directions are prioritized:

- 1. Audio-Visual Cross-Modal Synchronization: Integrating acoustic speech prosody features (via OpenAI Whisper or wav2vec 2.0) with visual lip-motion vectors to detect subtle temporal desynchronization between spoken phonemes and visual visemes.
- 2. Temporal Attention Modeling via Video Transformers: Augmenting frame-level predictions with TimeSformer or Video Vision Transformers (ViViT) to model inter-frame biological cadence, such as unnatural blinking frequencies and micro-expression anomalies.
- 3. Edge Optimization via TensorRT and ONNX Runtime: Quantizing model weights to INT8 precision using NVIDIA TensorRT to enable real-time (< 15ms) inference on low-power edge surveillance cameras, mobile devices, and automated border control kiosks.
- 4. Diffusion-Specific Spectral Forensics: Extending the frequency stream to model the unique high-frequency noise residuals characteristic of iterative denoising diffusion probabilistic models (DDPMs).

7.4 Adoption, Maintenance, and Forensic Rollout Plan

A structured rollout methodology is recommended for enterprise deployment: (1) Phase I (Pilot Integration): Deploy DeepGuard AI alongside existing manual KYC verification queues in fintech onboarding, operating in shadow-monitoring mode to collect baseline operational telemetry; (2) Phase II (Examiner Certification): Train digital forensic examiners to interpret dual Grad-CAM heatmaps and Radial Azimuthal profiles, establishing standard operating procedures (SOPs) for judicial testimony; and (3) Phase III (Automated Enforcement): Integrate the RESTful FastAPI inference endpoints directly into automated KYC gating systems, enabling real-time rejection of synthetic facial attacks with automated cryptographic certificate generation.

7.5 Prioritised Future Engineering Roadmap

The post-submission engineering roadmap is structured into three milestones: Milestone 1 (Months 1–3): Package the deep learning engine as an optimized Docker container with ONNX Runtime GPU acceleration; Milestone 2 (Months 4–6): Implement multi-face tracking for multi-subject group videos and crowd forensic analysis; Milestone 3 (Months 7–12): Submit the dual-stream architecture, empirical benchmarks, and spectral theory to a Scopus/IEEE indexed international journal on information forensics and cybersecurity.

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APPENDIX

Appendix A: Core Source Code Extracts

This appendix reproduces primary architectural source code components from the operational DeepGuard AI repository. The complete, version-controlled codebase is available at https://github.com/DevRanaKaila/DeepGuard-AI.

Code Listing: A.1 DualStreamNet Architecture (src/models/dual_stream_net.py)

```
class DualStreamNet(nn.Module):
    """Unified Dual-Stream Network combining Spatial and Frequency streams."""
    def __init__(self, pretrained: bool = True):
        super().__init__()
        self.spatial_stream = SpatialStream(pretrained=pretrained)
        self.frequency_stream = FrequencyStream()
        self.fusion = AttentionFusion(spatial_dim=1792, freq_dim=512, out_dim=1024)
        self.classifier = nn.Sequential(
            nn.Linear(1024, 256),
            nn.LayerNorm(256),
            nn.ReLU(inplace=True),
            nn.Dropout(0.4),
            nn.Linear(256, 1)
        )

    def forward(self, x_rgb: torch.Tensor, x_dct: torch.Tensor) -> Dict[str,
torch.Tensor]:
        v_spatial = self.spatial_stream(x_rgb)
        v_freq = self.frequency_stream(x_dct)
        v_fused, gate = self.fusion(v_spatial, v_freq)
        logits = self.classifier(v_fused)
        probs = torch.sigmoid(logits)
        return {
            'logits': logits,
            'probs': probs,
            'gate_weight': gate,
            'spatial_features': v_spatial,
            'freq_features': v_freq,
            'fused_features': v_fused
        }
```

Code Listing: A.2 4-Stage Spectral-CNN (src/models/frequency_stream.py)

```
class FrequencyStream(nn.Module):
    """4-Stage Custom Spectral-CNN for 2D-DCT Feature Extraction."""
    def __init__(self, in_channels: int = 1, feature_dim: int = 512):
        super().__init__()
        self.features = nn.Sequential(
            # Stage 1: Initial frequency receptive field
            nn.Conv2d(in_channels, 32, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            # Stage 2: Depthwise separable spectral filters
            nn.Conv2d(32, 64, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(64),
```

```
            nn.ReLU(inplace=True),
            # Stage 3: Mid-frequency feature maps
            nn.Conv2d(64, 128, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(128),
            nn.ReLU(inplace=True),
            # Stage 4: High-frequency artifact detection
            nn.Conv2d(128, 256, kernel_size=3, stride=2, padding=1),
            nn.BatchNorm2d(256),
            nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((1, 1))
        )
        self.fc = nn.Sequential(
            nn.Linear(256, feature_dim),
            nn.LayerNorm(feature_dim),
            nn.ReLU(inplace=True)
        )

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        x = self.features(x)
        x = torch.flatten(x, 1)
        return self.fc(x)
```

Appendix B: Additional Material & Project Metadata

B.1 Project Links and Version Control

- Official GitHub Source Repository: https://github.com/DevRanaKaila/DeepGuard-AI
- Local Interactive Forensic Dashboard: http://localhost:8501/
- Kaggle Benchmark Challenge Baseline: https://github.com/selimsef/dfdc_deepfake_challenge

B.2 Main Application Routes and CLI Commands

Table B.1: *Main application execution modes, CLI parameters, and operational commands*

Interface / Mode	Command / Route	Purpose and Target Subsystem
Web Dashboard	streamlit run app.py	Launches interactive Streamlit forensic analysis web application
Image Inference CLI	python inference.py --input <path> --output results	Executes single-image dual-stream inference with Grad-CAM
Video Inference CLI	python inference.py --input <video> --output results	Processes continuous video stream at 5 FPS with temporal aggregation

Model Training	python train.py --config config.py	Executes end-to-end model training with AdamW and label smoothing
Figure Generator	python generate_figures.py	Generates high-resolution architecture diagrams and evaluation plots

B.3 Repository Directory Structure

Code Listing: Repository Directory Structure (github.com/DevRanaKaila/DeepGuard-AI)

```
DeepGuard-AI/
├── src/
│   ├── data/                # Video decoding, face alignment & 2D-DCT transforms
│   │   ├── augmentation.py   # JPEG, blur, and color jitter augmentations
│   │   ├── dataset.py        # FaceForensics++ & Celeb-DF v2 data loaders
│   │   ├── dct_transform.py  # 2D-DCT-II & Radial Azimuthal Spectrum A(r)
│   │   └── preprocessing.py  # MTCNN face landmark tracking & crop extraction
│   ├── models/               # Neural network architectures
│   │   ├── dual_stream_net.py # Complete unified DualStreamNet model
│   │   ├── frequency_stream.py# 4-stage custom Spectral-CNN
│   │   ├── fusion.py         # Attention-Gated Bilinear Fusion module
│   │   ├── selim_classifier.py# Kaggle DFDC 1st-place champion backbone
│   │   └── spatial_stream.py # Pretrained EfficientNet-B4 spatial backbone
│   └── utils/                # Evaluation & visualization utilities
│       ├── gradcam.py        # Dual-stream Grad-CAM heatmap generator
│       ├── metrics.py        # AUC-ROC, EER, and confusion matrix evaluators
│       └── video_processor.py# Uniform temporal frame extraction engine
├── checkpoints/              # Model weights (.pth) & training checkpoints
├── demo/                     # Sample authentic & deepfake test videos / faces
├── results/                  # Generated forensic reports, heatmaps & JSON logs
├── report_assets/            # High-resolution architectural diagrams & plots
├── app.py                    # Interactive Streamlit forensic web application
├── inference.py              # Standalone CLI and API inference engine
├── train.py                  # PyTorch training pipeline with AMP & Cosine Annealing
├── config.py                 # Global hyperparameters and directory configurations
└── requirements.txt          # Production Python package dependencies
```